

A PENALTY APPROACH FOR NON-SELF-ADJOINT TOPOLOGY OPTIMIZATION PROBLEMS

WEI GONG AND YUANDA YE

ABSTRACT. We propose a novel penalty method framework for compliant mechanism problems, incorporating a convex nonlocal perimeter approximation scheme. We rigorously analyze the existence of solutions to the optimization problem derived from the penalty method. Furthermore, we establish that the discrete problem Γ -converges to the continuous problem, ensuring consistency across scales. To solve the discrete problem, we develop a projected gradient method that guarantees strict monotonic descent of the objective function. Numerical experiments on the compliant mechanism and heat dissipation problems validate the effectiveness of the proposed method, with results supported by finite element error estimates and convergence analysis. This framework provides a robust approach to addressing complex optimization challenges in computational mathematics with potential applications in engineering design.

1. PENALTY METHOD FOR NON-SELF-ADJOINT PROBLEMS

We first analyze the compliant mechanism problem to develop an effective convergent algorithm. The general approach is as follows:

- (1) Formulate the compliant mechanism problem as a penalty form;
- (2) Analyze the relationship between the penalty form compliant mechanism problem and the original problem;
- (3) Propose an efficient algorithm for the penalty form.

In the derivation of the first part, we omit the perimeter constraint, as it does not affect the derivation.

1.1. Transform into penalty form. Consider the following topology optimization problem

$$(1.1) \quad \min_{(\chi, u) \in \mathcal{U} \times H_D^1(\Omega)^d} J(\chi, u) = \int_{\Gamma_N} t_{out} \cdot u \, ds$$

subject to

$$(1.2) \quad \int_{\Omega} E(\chi) \varepsilon(u) : \varepsilon(\hat{u}) \, dx = \int_{\Gamma_N} t_{in} \cdot \hat{u} \, ds \quad \forall \hat{u} \in H_D^1(\Omega)^d,$$

where $E(\chi) := (\chi E_{\max} + (1 - \chi) E_{\min}) E_0$ for some constant tensor E_0 and

$$\mathcal{U} := \left\{ \chi \in BV(\Omega) : \chi \in \{0, 1\}, \int_{\Omega} \chi \, dx \leq \beta |\Omega| \right\}.$$

Introduce the adjoint problem: find $v \in H_D^1(\Omega)^d$ such that

$$(1.3) \quad \int_{\Omega} E(\chi) \varepsilon(v) : \varepsilon(\hat{v}) \, dx = \int_{\Gamma_N} t_{out} \cdot \hat{v} \, ds \quad \forall \hat{v} \in H_D^1(\Omega)^d.$$

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For any $u, v \in H_D^1(\Omega)^d$ we define the following bilinear and linear forms:

$$(1.4) \quad a(\chi; u, v) := \int_{\Omega} E(\chi) \varepsilon(u) : \varepsilon(v) \, dx,$$

$$(1.5) \quad \ell_{out}(v) := \int_{\Gamma_N} t_{out} \cdot v \, ds,$$

$$(1.6) \quad \ell_{in}(v) := \int_{\Gamma_N} t_{in} \cdot v \, ds.$$

Setting $\hat{u} = v$ in (1.2) and $\hat{v} = u$ in (1.3), we have

$$(1.7) \quad \ell_{in}(v) = a(\chi; u, v) = \ell_{out}(u).$$

Therefore, the optimization problem (1.1)-(1.2) can be transformed into the following form:

$$(1.8) \quad \min_{(\chi, u) \in \mathcal{U} \times H_D^1(\Omega)^d} J(\chi, u) = \ell_{out}(u) \quad \text{subject to} \quad a(\chi; u, \hat{u}) = \ell_{in}(\hat{u}) \quad \forall \hat{u} \in H_D^1(\Omega)^d.$$

Note that the constraint equation of the above optimization problem is the Euler-Lagrange equation of the following optimization problem:

$$u = \arg \min_{w \in H_D^1(\Omega)^d} \frac{1}{2} a(\chi; w, w) - \ell_{in}(w).$$

If we define

$$g(\chi, w) := \frac{1}{2} a(\chi; w, w) - \ell_{in}(w),$$

then it is obvious that $-\min_{w \in H_D^1(\Omega)^d} g(\chi, w) = \frac{1}{2} a(\chi; u, u)$ with u solving the state equation (1.2).

Let v be the solution to the adjoint state equation (1.3). To reformulate the optimization problem we introduce the following new variables

$$\sigma = E(\chi) \varepsilon(u), \quad \rho = E(\chi) \varepsilon(v).$$

In view of (1.7), we can transform the topology optimization problem into the following form [7]:

$$(1.9) \quad \begin{aligned} & \min_{(\chi, \sigma, \varrho) \in \mathcal{U} \times S} \tilde{J}(\chi, \sigma, \varrho) = \int_{\Omega} E^{-1}(\chi) \sigma : \varrho \, dx \\ & \text{subject to} \quad (\sigma, \varrho) = \arg \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \left(\int_{\Omega} E^{-1}(\chi) (\hat{\sigma} : \hat{\sigma} + \hat{\varrho} : \hat{\varrho}) \, dx \right), \end{aligned}$$

where $E^{-1}(\chi) := \left(\left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) \chi + \frac{1}{E_{\min}} \right) E_0^{-1}$, S is defined by

$$S := \{(\sigma, \varrho) \in (L^2(\Omega)^{d \times d})^2 : (\sigma, \varrho)_{i,j} = (\sigma, \varrho)_{j,i}, \nabla \cdot (\sigma, \varrho) = (0, 0), (\sigma, \varrho) \cdot \vec{n}|_{\Gamma_N} = (t_{in}, t_{out})\}.$$

In Section 1.2 we will prove that the reformulated problem (1.9) is equivalent to the original problem (1.8).

We utilize the penalty method from optimization theory to formulate a new optimization problem

$$(1.10) \quad \min_{(\chi, \sigma, \varrho) \in \mathcal{U} \times S} L(\chi, \sigma, \varrho) := \int_{\Omega} E^{-1}(\chi) \sigma : \varrho \, dx + \lambda \left(\tilde{g}(\chi, \sigma, \varrho) - \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho}) \right),$$

where λ is the penalty parameter, $\tilde{g}$ is defined by

$$\tilde{g}(\chi, \sigma, \varrho) := \int_{\Omega} E^{-1}(\chi) (\sigma : \sigma + \varrho : \varrho) \, dx.$$

We remark that χ , σ and ρ are independent optimization variables in the optimization problem (1.10), this is in contrast to the optimization problem (1.9). In Section 1.2 we will prove that in this specific setting, as long as λ is sufficiently large, the optimization problem in the penalty form (1.10) is equivalent to the optimization problem (1.9).

Of course, we could also derive an alternative penalty form starting from the original problem

$$(1.11) \quad \min_{(\chi, u, v) \in \mathcal{U} \times [H_D^1(\Omega)^d]^2} L(\chi) = \ell_{out}(u) + \lambda \left(g(\chi, u) + g(\chi, v) - \min_{(\hat{u}, \hat{v}) \in [H_D^1(\Omega)^d]^2} (g(\chi, \hat{u}) + g(\chi, \hat{v})) \right).$$

However, this penalty form (1.11) does not possess the property of being equivalent to the original problem (1.8) when λ is sufficiently large. Therefore, we will not discuss it in depth here. However, this

form has its advantages, as it can handle cases with body forces, which I have not seen addressed in the review articles on engineering problems that I consulted [10].

1.2. Equivalence between the original problem and the penalty form. First, we prove the following lemma, which reveals the equivalence between the minimization problem we introduced and the linear elasticity equation.

Lemma 1.1. *The first-order optimality condition for the minimization problem*

$$(1.12) \quad \min_{\sigma \in S_1} \int_{\Omega} E^{-1}(\chi) \sigma : \sigma \, dx$$

can be characterized as follows

$$(1.13) \quad \begin{cases} \sigma = E(\chi) \varepsilon(\lambda) & \text{in } \Omega, \\ \lambda = 0 & \text{on } \Gamma_D, \\ \lambda = -\mu & \text{on } \Gamma_N, \\ \nabla \cdot \sigma = 0 & \text{in } \Omega, \\ \sigma \cdot \vec{n} = t_{out} & \text{on } \Gamma_N, \end{cases}$$

where

$$S_1 := \{\sigma \in L^2(\Omega)^{d \times d} : \sigma_{i,j} = \sigma_{j,i}, \nabla \cdot \sigma = 0, \sigma \cdot \vec{n}|_{\Gamma_N} = t_{out}\}.$$

Proof. First, we introduce the function space

$$S_0 := \{\sigma \in L^2(\Omega)^{d \times d} : \sigma_{i,j} = \sigma_{j,i}\},$$

and construct the following Lagrange functional $L : S_0 \times H^1(\Omega) \times H^1(\Omega) \rightarrow \mathbb{R}$

$$\mathcal{L}(\sigma, \lambda, \mu) = \frac{1}{2} \int_{\Omega} E^{-1}(\chi) \sigma : \sigma \, dx + \int_{\Omega} (\nabla \cdot \sigma) \cdot \lambda \, dx + \int_{\Gamma_N} (\sigma \cdot \vec{n} - t_{out}) \cdot \mu \, ds.$$

Then we can derive the first-order optimality condition as follows:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda}(\hat{\lambda}) &= \int_{\Omega} \nabla \cdot \sigma \cdot \hat{\lambda} \, dx = 0, \\ \frac{\partial \mathcal{L}}{\partial \mu}(\hat{\mu}) &= \int_{\Gamma_N} (\sigma \cdot \vec{n} - t_{out}) \cdot \hat{\mu} \, ds = 0, \\ \frac{\partial \mathcal{L}}{\partial \sigma}(\hat{\sigma}) &= \int_{\Omega} E^{-1}(\chi) \sigma : \hat{\sigma} \, dx + \int_{\Omega} (\nabla \cdot \hat{\sigma}) \cdot \lambda \, dx + \int_{\Gamma_N} \hat{\sigma} \cdot \vec{n} \cdot \mu \, ds = 0, \end{aligned}$$

where $\hat{\lambda} \in H^1(\Omega)$, $\hat{\mu} \in H^1(\Omega)$ and $\hat{\sigma} \in S_0$. For the last equation, we have

$$\begin{aligned} 0 &= \int_{\Omega} E^{-1}(\chi) \sigma : \hat{\sigma} \, dx + \int_{\Omega} (\nabla \cdot \hat{\sigma}) \cdot \lambda \, dx + \int_{\Gamma_N} \hat{\sigma} \cdot \vec{n} \cdot \mu \, ds \\ &= \int_{\Omega} E^{-1}(\chi) \sigma : \hat{\sigma} \, dx + \int_{\Gamma_N \cup \Gamma_D} \hat{\sigma} \cdot \vec{n} \cdot \lambda \, dx - \int_{\Omega} \hat{\sigma} : \nabla \lambda \, dx + \int_{\Gamma_N} \hat{\sigma} \cdot \vec{n} \cdot \mu \, ds \\ &= \int_{\Omega} (E^{-1}(\chi) \sigma - \varepsilon(\lambda)) : \hat{\sigma} \, dx + \int_{\Gamma_D} \hat{\sigma} \cdot \vec{n} \cdot \lambda \, dx + \int_{\Gamma_N} \hat{\sigma} \cdot \vec{n} \cdot (\lambda + \mu) \, ds. \end{aligned}$$

Choosing arbitrary $\hat{\sigma} \in H_0^1(\Omega)$ we can derive that

$$E^{-1}(\chi) \sigma = \varepsilon(\lambda).$$

Setting $\hat{\sigma} \in H^1(\Omega)$ and $\hat{\sigma} \cdot \vec{n}|_{\Gamma_N} = 0$ implies

$$\lambda = 0 \quad \text{on } \Gamma_D.$$

Lastly, setting $\hat{\sigma} \in H^1(\Omega)$ yields

$$\lambda = -\mu \quad \text{on } \Gamma_N.$$

Thus, the result is proven, where λ and μ are Lagrange multipliers. $\square$

Here, we should note that problem (1.12) is a strictly convex optimization problem, so the solution is unique (if a solution exists), and the first-order optimality condition is both necessary and sufficient. On the other hand, the first-order optimality condition corresponds to the linear elasticity equation. Thus, we establish the existence and uniqueness of the solution to problem (1.12). This also demonstrates the existence and uniqueness of the solution to the optimization problem $\min_{(\hat{\sigma}, \hat{\varrho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho})$. Hence, we can prove the following theorem.

Theorem 1.2. *The problem (1.8) is equivalent to the problem (1.9), as they are different formulations of the same problem. Specifically, (1.8) is formulated with the displacement field as the state variable, while (1.9) is formulated with the stress field as the state variable.*

Proof. Let u^* and v^* be solutions of the state equation (1.2) and the adjoint state equation (1.3), respectively. We have established in Lemma 1.1 that the constraint of problem (1.9) is essentially equivalent to solving two linear elasticity equations, where $\sigma^* = E(\chi)\varepsilon(u^*)$ and $\varrho^* = E(\chi)\varepsilon(v^*)$ are the stress fields corresponding to the state equation (1.2) and the adjoint equation (1.3). Therefore, we have

$$\begin{aligned}\tilde{J}(\chi, \sigma^*, \varrho^*) &= \int_{\Omega} E^{-1}(\chi) \sigma^* : \varrho^* \, dx \\ &= \int_{\Omega} E(\chi) \varepsilon(u^*) : \varepsilon(v^*) \, dx \\ &= a(\chi; u^*, v^*) \\ &= \ell_{out}(u^*) \\ &= J(\chi, u^*),\end{aligned}$$

which demonstrates the equivalence between problem (1.8) and problem (1.9). $\square$

In the following, we prove the equivalence between the problem (1.9) and the penalty form (1.10).

Theorem 1.3. *Provided $\lambda > 1/2$, it holds that*

$$(1.14) \quad \min_{(\hat{\sigma}, \hat{\varrho}) \in S} L(\chi, \hat{\sigma}, \hat{\varrho}) = \tilde{J}(\chi, \sigma^*, \varrho^*),$$

where $(\sigma^*, \varrho^*) \in \arg \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho})$.

Proof. First, let u^* and v^* be the solutions of (1.2) and (1.3), respectively. We observe that

$$(1.15) \quad \sigma^* = E(\chi)\varepsilon(u^*), \quad \varrho^* = E(\chi)\varepsilon(v^*).$$

Next, we prove that when $\lambda > 1/2$, $L(\chi, \sigma^*, \varrho^*) = \min_{(\hat{\sigma}, \hat{\varrho}) \in S} L(\chi, \hat{\sigma}, \hat{\varrho})$. Note that

$$\begin{aligned}L(\chi, \hat{\sigma}, \hat{\varrho}) - L(\chi, \sigma^*, \varrho^*) &= \int_{\Omega} E^{-1}(\chi) \hat{\sigma} : \hat{\varrho} \, dx + \lambda \left(\int_{\Omega} E^{-1}(\chi) (\hat{\sigma} : \hat{\sigma} + \hat{\varrho} : \hat{\varrho}) \, dx \right) \\ &\quad - \lambda \left(\int_{\Omega} E^{-1}(\chi) (\sigma^* : \sigma^* + \varrho^* : \varrho^*) \, dx \right) - \int_{\Omega} E^{-1}(\chi) \sigma^* : \varrho^* \, dx \\ &= \frac{1}{2} \int_{\Omega} E^{-1}(\chi) (\hat{\sigma} + \hat{\varrho}) : (\hat{\sigma} + \hat{\varrho}) \, dx + \left(\lambda - \frac{1}{2} \right) \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho}) \\ &\quad - \frac{1}{2} \int_{\Omega} E^{-1}(\chi) (\sigma^* + \varrho^*) : (\sigma^* + \varrho^*) \, dx - \left(\lambda - \frac{1}{2} \right) \tilde{g}(\chi, \sigma^*, \varrho^*).\end{aligned}$$

Since $(\sigma^*, \varrho^*) \in \arg \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho})$, we only need to prove that

$$(\sigma^* + \varrho^*) \in \arg \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \int_{\Omega} E^{-1}(\chi) (\hat{\sigma} + \hat{\varrho}) : (\hat{\sigma} + \hat{\varrho}) \, dx.$$

Indeed, it follows from the fact $(\sigma^*, \varrho^*), (\hat{\sigma}, \hat{\varrho}) \in S$ that

$$\begin{aligned}&\int_{\Omega} E^{-1}(\chi) (\hat{\sigma} + \hat{\varrho}) : (\hat{\sigma} + \hat{\varrho}) \, dx - \int_{\Omega} E^{-1}(\chi) (\sigma^* + \varrho^*) : (\sigma^* + \varrho^*) \, dx \\ &= \int_{\Omega} E^{-1}(\chi) |\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*|^2 \, dx + 2 \int_{\Omega} E^{-1}(\chi) (\sigma^* + \varrho^*) : (\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*) \, dx \\ &= \int_{\Omega} E^{-1}(\chi) |\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*|^2 \, dx + 2 \int_{\Omega} (\varepsilon(u^*) + \varepsilon(v^*)) : (\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*) \, dx \\ &= \int_{\Omega} E^{-1}(\chi) |\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*|^2 \, dx - 2 \int_{\Omega} (u^* + v^*) \cdot \nabla \cdot (\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*) \, dx \\ &\quad + 2 \int_{\Gamma_D \cup \Gamma_N} (u^* + v^*) \cdot (\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*) \cdot \vec{n} \, ds \\ &= \int_{\Omega} E^{-1}(\chi) |\hat{\sigma} + \hat{\varrho} - \sigma^* - \varrho^*|^2 \, dx \geq 0.\end{aligned}$$

The inequality above indicates that we have identified a minimizer so that the problem has a solution.

Therefore, further discussion of the existence of solutions is unnecessary. This finishes the proof. $\square$

Therefore, we can immediately show that the compliant mechanism problem (1.8) is equivalent to the following double minimization problem:

$$(1.16) \quad \min_{\chi \in \mathcal{U}} \min_{(\sigma, \varrho) \in S} L(\chi, \sigma, \varrho).$$

Moreover, the inner minimization can be obtained by solving the state equation

$$(1.17) \quad a(\chi; u, \hat{u}) = \ell_{in}(\hat{u}) \quad \forall \hat{u} \in H_D^1(\Omega),$$

and the adjoint state equation

$$(1.18) \quad a(\chi; v, \hat{v}) = \ell_{out}(\hat{v}) \quad \forall \hat{v} \in H_D^1(\Omega).$$

1.3. Derivation of the algorithm. Since $\mathcal{U}$ is not a convex set, it is challenging to analyze the properties of the objective functional with respect to χ . To do this, we first relax χ to its convex hulls $\bar{\mathcal{U}}$ and consider the following double minimization problem

$$(1.19) \quad \min_{\chi \in \bar{\mathcal{U}}} \min_{(\sigma, \varrho) \in S} L(\chi, \sigma, \varrho) := \int_{\Omega} E^{-1}(\chi) \sigma : \varrho \, dx + \lambda \left(\tilde{g}(\chi, \sigma, \varrho) - \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho}) \right),$$

where

$$\bar{\mathcal{U}} := \left\{ \chi \in BV(\Omega) : \chi \in [0, 1], \int_{\Omega} \chi \, dx \leq \beta |\Omega| \right\}.$$

It should be noted that, due to the relaxation of χ from $\mathcal{U}$ to $\bar{\mathcal{U}}$, some previously valid equations no longer hold, such as:

$$(1.20) \quad a(\chi; u, v) = \int_{\Omega} E(\chi) \varepsilon(u) : \varepsilon(v) \, dx \neq \int_{\Omega} E^{-1}(\chi) \sigma : \varrho \, dx \quad \text{for some } \chi \in \bar{\mathcal{U}},$$

because $E^{-1}(\chi) \neq E(\chi)^{-1}$ when $\chi \in [0, 1]$. Therefore, for χ belonging to the closed convex set $\bar{\mathcal{U}}$, we introduce the following notation:

$$(1.21) \quad E(\chi) = \frac{1}{A(\chi)} E_0, \quad E^{-1}(\chi) = A(\chi) E_0^{-1},$$

and

$$A(\chi) = \left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) \chi + \frac{1}{E_{\min}}.$$

Clearly, when $\chi \in \mathcal{U}$ is the indicator function, equality is maintained

$$\frac{1}{A(\chi)} = (E_{\max} - E_{\min}) \chi + E_{\min}.$$

Therefore, these are reasonable extensions for $E^{-1}(\chi)$ and $E(\chi)$ defined in Section 1.1.

Throughout the remainder of this paper, we redefine $a(\chi; u, v) : L^\infty(\Omega) \times H_D^1(\Omega)^d \times H_D^1(\Omega)^d \rightarrow \mathbb{R}$ as follows

$$(1.22) \quad a(\chi; u, v) := \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(v) \, dx.$$

Of course, such an extension also introduces a problem, namely that the optimization problem formulated in terms of the stress field and the optimization problem formulated in terms of the displacement field are essentially two distinct problems, with different properties on $\mathcal{U}$. This is precisely the reason why transforming the problem to the stress field formulation yields better numerical performance. However, in the theoretical analysis section, we need to clearly articulate the relationship between each step of the transformation.

First, we derive the Fréchet derivative of L with respect to χ .

Lemma 1.4. *Define*

$$G(\chi) = \min_{(\hat{\sigma}, \hat{\varrho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\varrho}) := \int_{\Omega} A(\chi) E_0^{-1} (\hat{\sigma} : \hat{\sigma} + \hat{\varrho} : \hat{\varrho}) \, dx,$$

then from Lemma 1.1 we have

$$G(\chi) = \int_{\Omega} A(\chi) E_0^{-1} (\sigma(\chi) : \sigma(\chi) + \varrho(\chi) : \varrho(\chi)) \, dx.$$

Here we denote the stress fields corresponding to χ as $\sigma(\chi) := \frac{1}{A(\chi)} E_0 \varepsilon(u)$ and $\varrho(\chi) := \frac{1}{A(\chi)} E_0 \varepsilon(v)$ to emphasize their dependence on χ , where $u \in H_D^1(\Omega)^d$ and $v \in H_D^1(\Omega)^d$ satisfy

$$(1.23) \quad \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(\hat{u}) \, dx = \int_{\Gamma_N} t_{in} \cdot \hat{u} \, ds \quad \forall \hat{u} \in H_D^1(\Omega)^d$$

and

$$(1.24) \quad \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(v) : \varepsilon(\hat{v}) \, dx = \int_{\Gamma_N} t_{out} \cdot \hat{v} \, ds \quad \forall \hat{v} \in H_D^1(\Omega)^d.$$

Moreover, there holds

$$\frac{\partial G}{\partial \chi}(\chi)(\hat{\chi}) = \int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1}(\sigma(\chi) : \sigma(\chi) + \varrho(\chi) : \varrho(\chi)) \, dx.$$

Proof. From the definition of $\sigma(\chi)$ and $\varrho(\chi)$, $G(\chi)$ can be transformed into

$$(1.25) \quad G(\chi) = \int_{\Omega} \frac{1}{A(\chi)} E_0(\varepsilon(u) : \varepsilon(u) + \varepsilon(v) : \varepsilon(v)) \, dx.$$

Since the two components of $G(\chi)$ have identical forms, we simply consider the following functional

$$j(\chi) = \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(u) \, dx$$

subject to

$$\int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(\hat{u}) \, dx = \int_{\Gamma_N} t_{in} \cdot \hat{u} \, ds \quad \forall \hat{u} \in H_D^1(\Omega)^d.$$

Hence, it's Lagrange functional reads as follows

$$\ell(\chi, u, \hat{u}) = \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(u) \, dx + \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(\hat{u}) \, dx - \int_{\Gamma_N} t_{in} \cdot \hat{u} \, ds.$$

It is straightforward to show that $\hat{u} = -2u$. Therefore,

$$\frac{\partial j}{\partial \chi}(\chi)(\hat{\chi}) = \int_{\Omega} \frac{A'(\chi)\hat{\chi}}{A(\chi)^2} E_0 \varepsilon(u) : \varepsilon(u) \, dx.$$

Furthermore, since $\sigma(\chi) = \frac{1}{A(\chi)} E_0 \varepsilon(u)$, it follows that

$$\frac{\partial j}{\partial \chi}(\chi)(\hat{\chi}) = \int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1} \sigma(\chi) : \sigma(\chi) \, dx.$$

Hence,

$$\begin{aligned} \frac{\partial G}{\partial \chi}(\chi)(\hat{\chi}) &= \int_{\Omega} \frac{A'(\chi)(\hat{\chi})}{A(\chi)^2} E_0(\varepsilon(u) : \varepsilon(u) + \varepsilon(v) : \varepsilon(v)) \, dx \\ &= \int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1}(\sigma(\chi) : \sigma(\chi) + \varrho(\chi) : \varrho(\chi)) \, dx. \end{aligned}$$

This finishes the proof. $\square$

From Lemma 1.4, the Fréchet derivative of L is directly obtained as follows:

$$(1.26) \quad \begin{aligned} \frac{\partial L}{\partial \chi}(\chi, \sigma, \varrho)(\hat{\chi}) &= \int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1} \sigma : \varrho \, dx + \lambda \left(\int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1}(\sigma : \sigma + \varrho : \varrho) \, dx \right. \\ &\quad \left. - \int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1}(\sigma(\chi) : \sigma(\chi) + \varrho(\chi) : \varrho(\chi)) \, dx \right). \end{aligned}$$

In particular, if $(\sigma, \varrho) = (\sigma(\chi), \varrho(\chi))$, then

$$(1.27) \quad \frac{\partial L}{\partial \chi}(\chi, \sigma(\chi), \varrho(\chi))(\hat{\chi}) = \int_{\Omega} A'(\chi)(\hat{\chi}) E_0^{-1} \sigma(\chi) : \varrho(\chi) \, dx.$$

Next, we demonstrate that $\min_{(\sigma, \varrho) \in S} \tilde{g}(\chi, \sigma, \varrho)$ is concave with respect to χ . We only need to prove the following lemma.

Lemma 1.5. *The functional*

$$g_1(\chi) = \min_{\sigma \in S_1} \int_{\Omega} E^{-1}(\chi) \sigma : \sigma \, dx$$

is concave with respect to χ .

103 *Proof.* It follows from the proof of Lemma 1.4 that

$$g_1(\chi) = \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(u) \, dx,$$

104 where u solves the state equation (1.23). Here, we do not employ the Lagrangian approach but instead
 105 use the implicit function theorem. Denoting $u' = \frac{\partial u}{\partial \chi}(\hat{\chi})$ and $u'' = \frac{\partial^2 u}{\partial \chi^2}(\hat{\chi}^2)$, by the chain rule we can
 106 compute

$$\begin{aligned} \frac{\partial^2 g_1}{\partial \chi^2}(\hat{\chi}^2) = & 2 \int_{\Omega} \frac{[A'(\chi)(\hat{\chi})]^2}{A(\chi)^3} E_0 \varepsilon(u) : \varepsilon(u) \, dx - 4 \int_{\Omega} \frac{A'(\chi)(\hat{\chi})}{A(\chi)^2} E_0 \varepsilon(u) : \varepsilon(u') \, dx \\ & + 2 \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u') : \varepsilon(u') \, dx + 2 \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u) : \varepsilon(u'') \, dx. \end{aligned}$$

107 Differentiating with respect to χ in (1.23) we obtain

$$- \int_{\Omega} \frac{A'(\chi)(\hat{\chi})}{A(\chi)^2} E_0 \varepsilon(u) : \varepsilon(\hat{u}) \, dx + \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u') : \varepsilon(\hat{u}) \, dx = 0,$$

108

$$2 \int_{\Omega} \frac{[A'(\chi)(\hat{\chi})]^2}{A(\chi)^3} E_0 \varepsilon(u) : \varepsilon(\hat{u}) \, dx - 2 \int_{\Omega} \frac{A'(\chi)(\hat{\chi})}{A(\chi)^2} E_0 \varepsilon(u') : \varepsilon(\hat{u}) \, dx + \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u'') : \varepsilon(\hat{u}) \, dx = 0$$

109 for any $\hat{u} \in H_D^1(\Omega)^d$. Hence, setting $\hat{u} = u'$ in the first equation yields

$$(1.28) \quad \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u') : \varepsilon(u') \, dx = \int_{\Omega} \frac{A'(\chi)(\hat{\chi})}{A(\chi)^2} E_0 \varepsilon(u) : \varepsilon(u') \, dx,$$

110 while setting $\hat{u} = u$ in the second equation gives

$$\int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u'') : \varepsilon(\hat{u}) \, dx = -2 \int_{\Omega} \frac{[A'(\chi)(\hat{\chi})]^2}{A(\chi)^3} E_0 \varepsilon(u) : \varepsilon(u) \, dx + 2 \int_{\Omega} \frac{A'(\chi)(\hat{\chi})}{A(\chi)^2} E_0 \varepsilon(u') : \varepsilon(u) \, dx.$$

111 By applying the Cauchy-Schwarz inequality in (1.28), we obtain

$$(1.29) \quad \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u') : \varepsilon(u') \, dx \leq \int_{\Omega} \frac{[A'(\chi)(\hat{\chi})]^2}{A(\chi)^3} E_0 \varepsilon(u) : \varepsilon(u) \, dx.$$

112 Therefore

$$\frac{\partial^2 g_1(\chi, u, v)}{\partial \chi^2}(\hat{\chi})(\hat{\chi}) = -2 \int_{\Omega} \frac{[A'(\chi)(\hat{\chi})]^2}{A(\chi)^3} E_0 \varepsilon(u) : \varepsilon(u) \, dx + 2 \int_{\Omega} \frac{1}{A(\chi)} E_0 \varepsilon(u') : \varepsilon(u') \, dx \leq 0.$$

113 This finishes the proof. $\square$

114 Since $\tilde{J}$ and $\tilde{g}$ are linear functionals with respect to χ , it follows from Lemma 1.5 that $L(\chi, \sigma, \rho) =$
 115 $\tilde{J}(\chi, \sigma, \rho) + \lambda(\tilde{g}(\chi, \sigma, \rho) - \min_{(\hat{\sigma}, \hat{\rho}) \in S} \tilde{g}(\chi, \hat{\sigma}, \hat{\rho}))$ is convex with respect to χ .

116 To ensure the existence of solutions, we need to add a perimeter penalty term to L . We consider the
 117 regularized problem

$$(1.30) \quad L^\gamma(\chi, \sigma, \varrho) = L(\chi, \sigma, \varrho) + \gamma \|\chi\|_{TV},$$

118 where $\|\cdot\|_{TV}$ represents the total variation norm in the space of functions with bounded variation. We
 119 remark that the TV-norm is computationally challenging, so one usually uses an approximate perimeter
 120 regularization for the implementation. To maintain the convexity of the objective functional, we introduce
 121 a perimeter approximation, resulting in the following approximate objective functional:

$$(1.31) \quad L^{\gamma, \epsilon}(\chi, \sigma, \varrho) = L(\chi^\epsilon, \sigma, \varrho) + \frac{\gamma}{\epsilon} P_{G_\epsilon}(\chi),$$

122 where $\chi^\epsilon := G_\epsilon * \chi$ and the family $\{G_\epsilon \in C_c^\infty(\mathbb{R}^d)\}_\epsilon$ is a Dirac sequence.

123 Specifically, we assume that G is a smooth kernel satisfying $\int_{\mathbb{R}^d} G(x) \, dx = 1$. We define $G_\epsilon(x) =$
 124 $\frac{1}{\epsilon^d} G\left(\frac{x}{\epsilon}\right)$, which is introduced for the need of theoretical analysis, and

$$(1.32) \quad P_{G_\epsilon}(\chi) := \frac{1}{2} \int_{\Omega} \int_{\Omega} G_\epsilon(x - y) |\chi(x) - \chi(y)| \, dx dy.$$

It can be shown that the nonlocal convex approximation $\frac{C}{\epsilon}P_{G_\epsilon}(\chi)$, with $C > 0$ a constant, Γ -converges to $\|\chi\|_{TV}$, we refer to [9] for more details. We can provide a subderivative of this functional as follows:

$$\partial P_{G_\epsilon}(\chi) = \left\{ g \in L^\infty(\Omega) \mid \langle g, \tilde{\chi} \rangle = \int_\Omega \left(\int_\Omega G_\epsilon(x-y) \xi(x,y) dy \right) \tilde{\chi}(x) dx, \xi(x,y) \in \text{sign}(\chi(x) - \chi(y)) \right\}.$$

Consequently, we obtain an optimization problem P that can be executed efficiently

$$\min_{(\chi, \sigma, \varrho) \in \mathcal{U} \times S} L^{\gamma, \epsilon}(\chi, \sigma, \varrho) = \int_\Omega A(\chi^\epsilon) E_0^{-1} \sigma : \varrho dx + \lambda \left(\int_\Omega A(\chi^\epsilon) E_0^{-1} (|\sigma|^2 + |\varrho|^2 - |\sigma^*|^2 - |\varrho^*|^2) dx \right) + \frac{\gamma}{\epsilon} P_{G_\epsilon}(\chi).$$

We emphasize again that the inner minimizer of this double minimization problem can be obtained by solving the corresponding linear elasticity equation, while the outer minimization is a nonconvex optimization problem. However, we have established that the objective functional $L^{\gamma, \epsilon}$ is convex on $\bar{\mathcal{U}}$. Therefore, we adopt the approach of combining the projected gradient with a proximal point algorithm.

Let χ_k be the current iteration point and (σ_k, ϱ_k) be the solution to the corresponding inner minimization problem. We first perform a proximal point update in $L^\infty(\Omega)$:

$$\bar{\chi}_{k+1} = \arg \min_{\chi \in L^\infty(\Omega)} \left\{ L^{\gamma, \epsilon}(\chi_k, \sigma_k, \varrho_k) + \left\langle \frac{\partial L^{\gamma, \epsilon}}{\partial \chi}(\chi_k, \sigma_k, \varrho_k), \chi - \chi_k \right\rangle + \frac{1}{2r} \|\chi - \chi_k\|_{0, \Omega}^2 \right\}.$$

Then, we project the resulting minimizer $\bar{\chi}_{k+1}$ onto $\mathcal{U}$

$$\chi_{k+1} = \mathcal{P}_{\mathcal{U}}(\bar{\chi}_{k+1}).$$

In the following, we derive the explicit computational forms for the update of the proximal point (1.35) and the projection (1.36). First, for the update of the proximal point, the key lies in the selection of the subgradient. Here, we choose

$$\langle g_k, \tilde{\chi} \rangle = \int_\Omega \left(\int_\Omega G_\epsilon(x-y) (\chi_k(x) - \chi_k(y)) dy \right) \tilde{\chi}(x) dx,$$

so that $g_k \in \partial P_{G_\epsilon}(\chi)$. Recalling (1.27), the calculation proceeds as follows:

$$\begin{aligned} & \left\langle \frac{\partial L^{\gamma, \epsilon}}{\partial \chi}(\chi_k, \sigma_k, \varrho_k), \chi - \chi_k \right\rangle \\ &= \left\langle \frac{\partial L(\chi_k^\epsilon, \sigma_k, \varrho_k)}{\partial \chi}, \chi - \chi_k \right\rangle + \frac{\gamma}{\epsilon} \langle g_k, \chi - \chi_k \rangle \\ &= \int_\Omega A'(G_\epsilon * (\chi - \chi_k)) E_0^{-1} \sigma_k : \varrho_k dx + \frac{\gamma}{\epsilon} \int_\Omega \left(\int_\Omega G_\epsilon(x-y) (\chi_k(x) - \chi_k(y)) dy \right) (\chi(x) - \chi_k(x)) dx \\ &= \int_\Omega (\chi - \chi_k) \left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) G_\epsilon * (E_0^{-1} \sigma_k : \varrho_k) dx + \frac{\gamma}{\epsilon} \int_\Omega (\chi - \chi_k) (\chi_k - G_\epsilon * \chi_k) dx. \end{aligned}$$

The last equality is obtained by extending the function to zero and then using the properties of convolution. Hence, we have

$$\bar{\chi}_{k+1} = \chi_k - r \left(\left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) G_\epsilon * (E_0^{-1} \sigma_k : \varrho_k) + \frac{\gamma}{\epsilon} (\chi_k - G_\epsilon * \chi_k) \right).$$

Next, we address the computational issue of the projection. The key aspect of the projection lies in the choice of norm under which the projection is defined, as different norms lead to different projection results, directly impacting our study of convergence. Here, we use the L^1 norm.

It is easy to see that for any L^p norm, the projection result is an element defined as follows: Let $c_{k+1} = \inf\{c \in \mathbb{R} : \mu(\{x : \bar{\chi}_{k+1}(x) > c\}) \leq \beta|\Omega|\}$, then

$$\chi_{k+1}(x) = \begin{cases} 1 & \text{if } \bar{\chi}_{k+1}(x) > c_{k+1}, \\ 0 & \text{if } \bar{\chi}_{k+1}(x) \leq c_{k+1}, \end{cases}$$

where μ is a measure of $\mathbb{R}^d$.

With the update formulas (1.37) and (1.38), we can easily obtain an estimate χ_{k+1} . Since our objective functional is convex, we can view the proximal point algorithm as a second-order approximation, where the size of the proximal factor r determines the reliability of the second-order approximation. However, we do not know the optimal value of r . Therefore, we need to adjust the size of r to obtain a suitable χ_{k+1} . Here, we do not adopt the classical Wolfe conditions, as the outer optimization problem at the discrete

level is a 0-1 programming problem. It is sufficient to ensure that the objective function decreases. Thus, we use a binary search to find an appropriate r , thereby completing an iteration step.

Clearly, the choice of the search interval for r significantly affects the number of searches, which in turn impacts the algorithm's efficiency. By carefully examining the update and projection steps of the proximal point algorithm, we observe that if the update does not alter the relative ordering of the points where χ_k takes values 1 and 0, then $\bar{\chi}_{k+1} = \chi_k$. Consequently, we can choose

$$r_{\min} = \frac{1}{2} \sup_{x \in \Omega} \left(\left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) G_{\epsilon} * (E_0^{-1} \sigma_k : \varrho_k) + \frac{\gamma}{\epsilon} (\chi_k - G_{\epsilon} * \chi_k) \right)^{-1},$$

while the choice of $r_{\max}$ is relatively arbitrary, as long as $r_{\max} > r_{\min}$. The $\bar{\chi}_{k+1}$ obtained from $r_{\max}$ falls into one of the following three cases:

- (1) If $\bar{\chi}_{k+1}$ reduces the objective function, the search can be terminated, and we set $\chi_{k+1} \leftarrow \bar{\chi}_{k+1}$;
- (2) If $\bar{\chi}_{k+1} = \chi_k$, we update $r_{\min} \leftarrow r_{\max}$ and select a larger $r_{\max}$;
- (3) If $\bar{\chi}_{k+1}$ increases the objective function, it indicates that $r_{\max}$ is too large, and we can perform a line search within this interval.

1.4. Numerical implementation of the algorithm. In the following, we present the numerical implementation of the algorithm in pseudo-code form.

Algorithm 1 The coefficient line search algorithm

```

1: Initialize  $r_{\min}, r_{\max}, \Delta$ , and let  $r = \frac{r_{\max} + r_{\min}}{2}$ .
2: while 1 do
3:   Compute  $\bar{\chi}$  using (1.37).
4:   Compute  $\bar{\chi}_{k+1}$  using (1.38).
5:   Compute  $L^{\gamma, \epsilon}(\bar{\chi}_{k+1}, \sigma_k, \varrho_k)$ 
6:
7:   if  $L^{\gamma, \epsilon}(\bar{\chi}_{k+1}, \sigma_k, \varrho_k) > L^{\gamma, \epsilon}(\chi_k, \sigma_k, \varrho_k)$  then
8:      $r_{\max} \leftarrow r$ .
9:
10:    if  $\|\bar{\chi}_{k+1} - \chi_k\|_1 \leq \Delta$  then
11:       $\chi_{k+1} \leftarrow \chi_k$ , break;
12:
13:    end if
14:
15:    else if  $L^{\gamma, \epsilon}(\bar{\chi}_{k+1}, \sigma_k, \varrho_k) < L^{\gamma, \epsilon}(\chi_k, \sigma_k, \varrho_k)$  then
16:       $\chi_{k+1} \leftarrow \bar{\chi}_{k+1}$ , break;
17:
18:    else if  $L^{\gamma, \epsilon}(\bar{\chi}_{k+1}, \sigma_k, \varrho_k) = L^{\gamma, \epsilon}(\chi_k, \sigma_k, \varrho_k)$  and  $\|\bar{\chi}_{k+1} - \chi_k\|_1 > \Delta$  then
19:       $r_{\max} \leftarrow r$ .
20:
21:    end if
22:
23:    if  $\bar{\chi}_{k+1} == \chi^k$  then
24:       $r_{\min} \leftarrow r$ .
25:
26:    end if
27:     $r = \frac{r_{\max} + r_{\min}}{2}$ .
28: end while
29: return New graph  $\chi_{k+1}$  and function value  $L^{\gamma, \epsilon}(\chi_{k+1}, \sigma_k, \varrho_k)$ .
```

It is noteworthy that we introduce a parameter Δ here, which serves to terminate the line search and return χ_k when a change of χ_k by Δ (in the L^1 sense) along the descent direction leads to an increase in the objective function value. This effectively terminates the entire optimization algorithm. At the discrete level, we can choose $\Delta = C\mu(\Delta\Omega)$, where $\Delta\Omega$ represents the element mesh.

Algorithm 2 Project proximal algorithm for the compliant mechanisms problem

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- 1: Initialize $k = 0, \chi_k$ (The initial state is typically set to a uniform distribution with $\chi_k = \beta$), and parameter $\lambda, \gamma, \epsilon, \Delta$.
 - 2: **while** χ_k not converged **do**
 - 3: Solve $a(\chi_k^\epsilon; u, \hat{u}) = \ell_{in}(\hat{u}), \quad \forall \hat{u} \in H_D^1(\Omega)^d$ and $a(\chi_k^\epsilon; v, \hat{v}) = \ell_{out}(\hat{v}), \quad \forall \hat{v} \in H_D^1(\Omega)^d$.
 - 4: Compute $\left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}}\right) G_\epsilon * (E_0^{-1} \sigma_k : \varrho_k) + \frac{\gamma}{\epsilon} (\chi_k - G_\epsilon * \chi_k)$.
 - 5: Update χ_k using Algorithm 1.
 - 6: $k \leftarrow k + 1$.
 - 7: **end while**
 - 8: **return** Optimal graph χ^* and function value $L^{\gamma, \epsilon}(\chi^*, \sigma^*, \varrho^*)$.
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2. THEORETICAL ANALYSIS OF CONVERGENCE

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In this section, we investigate the convergence of such methods from an abstract perspective. First, we standardize the notation: variables with subscript n represent those in the discrete problem, which vary with the grid size h_n , while variables without subscript n denote those in the continuous problem. To clearly present our analysis process and for the convenience of notation, we restate the three optimization problems. We introduce the notation $X_n = \mathcal{U}_n \times S_n$ and $X = \mathcal{U} \times S$, where $\mathcal{U}_n \subset \mathcal{U}$ consists of piecewise constant finite element functions and S_n denotes the finite element space that is conforming.

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2.1. Problem description. The origin problem Q is defined as follows:

$$(2.1) \quad \min_{(\chi, \sigma, \varrho) \in X} L^{\gamma, 0}(\chi, \sigma, \varrho) = L^0(\chi, \sigma, \varrho) + \frac{\gamma}{C_G} \|\chi\|_{TV},$$

where $C_G := \lim_{\epsilon \downarrow 0} \frac{\|\chi\|_{TV}}{P_{G_\epsilon}(\chi)/\epsilon}$, and

$$L^0(\chi, \sigma, \varrho) := \int_{\Omega} A(\chi) E_0^{-1} \sigma : \varrho \, dx + \lambda \left(\int_{\Omega} A(\chi) E_0^{-1} (|\sigma|^2 + |\varrho|^2 - |\sigma^*|^2 - |\varrho^*|^2) \, dx \right).$$

Here $(\sigma^*, \varrho^*) := (\frac{1}{A(\chi)} E_0 \varepsilon(u^*), \frac{1}{A(\chi)} E_0 \varepsilon(v^*))$ where

$$(u^*, v^*) = \arg \min_{(u, v) \in [H_D^1(\Omega)^d]^2} \left(\frac{1}{2} (a(\chi; u, u) + a(\chi; v, v) - \ell_{in}(u) - \ell_{out}(v)) \right).$$

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The functions (σ, ϱ) are independent of χ while (σ^*, ϱ^*) depend on χ (in our application, they are solutions to PDEs).

The continuous problem P is given by

$$(2.2) \quad \min_{(\chi, \sigma, \varrho) \in X} L^{\gamma, \epsilon_k}(\chi, \sigma, \varrho) = L^{\epsilon_k}(\chi^{\epsilon_k}, \sigma, \varrho) + \frac{\gamma}{\epsilon_k} P_{G_{\epsilon_k}}(\chi),$$

where $\chi^{\epsilon_k} := G_{\epsilon_k} * \chi$, $\epsilon_k \rightarrow 0$ as $k \rightarrow +\infty$ and

$$L^{\epsilon_k}(\chi^{\epsilon_k}, \sigma, \varrho) := \int_{\Omega} A(\chi^{\epsilon_k}) E_0^{-1} \sigma : \varrho \, dx + \lambda \left(\int_{\Omega} A(\chi^{\epsilon_k}) E_0^{-1} (|\sigma|^2 + |\varrho|^2 - |\sigma_k^*|^2 - |\varrho_k^*|^2) \, dx \right).$$

Here $(\sigma_k^*, \varrho_k^*) := (\frac{1}{A(\chi^{\epsilon_k})} E_0 \varepsilon(u_k^*), \frac{1}{A(\chi^{\epsilon_k})} E_0 \varepsilon(v_k^*))$ where

$$(u_k^*, v_k^*) = \arg \min_{(u, v) \in [H_D^1(\Omega)^d]^2} \left(\frac{1}{2} (a(\chi^{\epsilon_k}; u, u) + a(\chi^{\epsilon_k}; v, v) - \ell_{in}(u) - \ell_{out}(v)) \right).$$

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The discrete problem P_n is defined as follows:

$$(2.3) \quad \min_{(\chi_n, \sigma_n, \varrho_n) \in X_n} L_n^{\gamma, \epsilon_k}(\chi_n, \sigma_n, \varrho_n) = L_n^{\epsilon_k}(\chi_n^{\epsilon_k}, \sigma_n, \varrho_n) + \frac{\gamma}{\epsilon_k} P_{G_{\epsilon_k}}(\chi_n),$$

where

$$L_n^{\epsilon_k}(\chi_n^{\epsilon_k}, \sigma_n, \varrho_n) := \int_{\Omega} A(\chi_n^{\epsilon_k}) E_0^{-1} \sigma_n : \varrho_n \, dx + \lambda \left(\int_{\Omega} A(\chi_n^{\epsilon_k}) E_0^{-1} (|\sigma_n|^2 + |\varrho_n|^2 - |\sigma_n^*|^2 - |\varrho_n^*|^2) \, dx \right).$$

Here $(\sigma_n^*, \varrho_n^*) := (\frac{1}{A(\chi^{\epsilon_k})} E_0 \varepsilon(u_n^*), \frac{1}{A(\chi^{\epsilon_k})} E_0 \varepsilon(v_n^*))$ where

$$(u_n^*, v_n^*) = \arg \min_{(u,v) \in S_n \times S_n} \left(\frac{1}{2} (a(\chi^{\epsilon_k}; u, u) + a(\chi^{\epsilon_k}; v, v) - \ell_{in}(u) - \ell_{out}(v)) \right).$$

178 The functions σ_n^* and ϱ_n^* depend on $\chi_n^{\epsilon_k}$ and thus χ_n (in our application, they are FEM solutions of
179 PDEs).

180 **2.2. Mathematical tools.** In this subsection we collect some known results from literature which will
181 be used in our following convergence analysis.

182 **Definition 2.1** ([3, Definition 2.3, Chapter 2, Page 9]). The characterization of Γ -convergence is as
183 follows:

- (1) **Liminf inequality (lower semicontinuity):** For every $u \in X$ and every sequence $\{u_n\} \subset X$ such that $u_n \rightarrow u$ in X ,

$$\liminf_{n \rightarrow +\infty} L_n(u_n) \geq L(u).$$

- (2) **Limsup inequality (recovery sequence):** For every $u \in X$, there exists a sequence $\{u_n\} \subset X$ such that $u_n \rightarrow u$ in X and

$$\limsup_{n \rightarrow +\infty} L_n(u_n) \leq L(u).$$

Definition 2.2. Let $\{f_n\}, f : \mathcal{X} \rightarrow \mathcal{Y}$. We say that f_n converges continuously to f , denoted by $f_n \xrightarrow{c} f$, if for any $x_n \xrightarrow{\mathcal{X}} x$, one has

$$\|f_n(x_n) - f(x)\|_{\mathcal{Y}} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Definition 2.3. Let $\{f_n : \mathcal{X}_n \rightarrow \mathcal{Y}\}$ for $\mathcal{X}_n \subset \mathcal{X}$ and $f : \mathcal{X} \rightarrow \mathcal{Y}$. We say that f_n sub-continuously converges to f , denoted by $f_n \xrightarrow{sub-c} f$, if for any $x_n \in \mathcal{X}_n \xrightarrow{\mathcal{X}} x$, one has

$$\|f_n(x_n) - f(x)\|_{\mathcal{Y}} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

184 **Lemma 2.4** ([2, Lemma 1.4, Chapter 1, Page 5]). For any $f \in L^q(\Omega)$ and $g \in L^r(\Omega)$, we have the
185 following Young's convolution inequality

$$\|f * g\|_{0,p,\Omega} \leq \|f\|_{0,q,\Omega} \|g\|_{0,r,\Omega},$$

186 where $p, q, r \in [1, +\infty]$ satisfy $\frac{1}{p} + 1 = \frac{1}{q} + \frac{1}{r}$.

187 **Lemma 2.5** ([5, Theorem 4.22, Chapter 4, Page 109]). Let $\rho_\epsilon(x) = \epsilon^{-d} \rho(x/\epsilon)$ be a mollifier in
188 $\mathbb{R}^d$, where $\rho \in C_c^\infty(\mathbb{R}^d)$, $\rho \geq 0$, $\int_{\mathbb{R}^d} \rho(x) dx = 1$, and ρ is radial (e.g., a Gaussian kernel $\rho(x) =$
189 $(2\pi)^{-d/2} \exp(-|x|^2/2)$). For any $f \in L^p(\mathbb{R}^d)$, $1 \leq p < \infty$, the convolution $\rho_\epsilon * f$ converges to f in
190 $L^p(\mathbb{R}^d)$:

$$\|\rho_\epsilon * f - f\|_{L^p(\mathbb{R}^d)} \rightarrow 0 \quad \text{as } \epsilon \rightarrow 0.$$

191 This result extends to bounded domains $\Omega \subset \mathbb{R}^d$ with Lipschitz boundary by zero extension of f to $\mathbb{R}^d$.

Lemma 2.6 ([8, Theorem 5.2, Chapter 5, Page 199]). Suppose $f_k \in BV(\Omega)$ ($k = 1, 2, \dots$) and

$$f_k \rightarrow f \quad \text{in } L_{loc}^1(\Omega).$$

Then we have the following lower semicontinuity of variation measure

$$\lim_{k \rightarrow \infty} \|f_k\|_{TV} \geq \|f\|_{TV}.$$

Lemma 2.7 ([9, Page 238]). The approximation $\frac{C_G}{\epsilon} P_{G_\epsilon}(\cdot)$ Γ -converges and pointwise converges to $\|\cdot\|_{TV}$. For each $\chi \in BV(\Omega)$, we have (cf. [9])

$$\lim_{\epsilon \downarrow 0} \frac{C}{\epsilon} P_{G_\epsilon}(\chi) = \|\chi\|_{TV},$$

where

$$C_G = \frac{2}{\int_{\Omega} G_1(x) |x_d| dx}.$$

192 **Definition 2.8.** Let (X, τ) be a topological space and $\{f_n\}$ be a sequence of functions such that $f_n : X \rightarrow \mathbb{R} \cup \{+\infty\}$. The sequence $\{f_n\}$ is said to be *equicoercive* if, for every $t \in \mathbb{R}$, there exists a compact
193 set $K_t \subset X$ such that, for all n , the sublevel set $\{x \in X : f_n(x) \leq t\} \subset K_t$.
194

Lemma 2.9 ([3, Theorem 5.1, Chapter 5, Page 67]). *Let (X, d) be a metric space and $f_n, f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a sequence of functions such that each f_n is coercive and lower semicontinuous, and the sequence f_n Γ -converges to f and is equicoercive. Suppose $x^* \in X$ is an isolated local minimizer of f , i.e., there exists a neighborhood $U \subset X$ of x^* such that:*

- (1) $f(x^*) < f(x)$ for all $x \in U$.
- (2) x^* is the only local minimizer in U .

If $\{x_n\} \subset U$ is a sequence of local minimizers of f_n such that $x_n \rightarrow x$ in the metric d , then $x = x^$.*

2.3. Relationships among the three optimization problems. In this subsection we will establish the connections among the three optimization problems Q , P and P_n . The proof strategy is as follows. First, we demonstrate that problem P_n Γ -converges to problem Q , and discuss the relationships among the three optimization problems. Then, we use Lemma 2.9 to prove that the local minimizers of P_n converge to the isolated local minimizers of Q . Finally, we prove that the solution sequence generated by optimization algorithm for solving P_n converges to the local minimizers of P_n .

Lemma 2.10. *If the domain Ω has a Lipschitz boundary Γ , then the origin problem Q has at least one solution.*

Proof. Let $\{\chi_k\} \subset \mathcal{U}$ be a minimizing sequence for the problem Q . Thus, $\{\chi_k\} \subset BV(\Omega)$. Since $BV(\Omega) \hookrightarrow L^1(\Omega)$, there exists a convergent subsequence (still denoted by $\{\chi_k\}$) that converges to some χ^* in $L^1(\Omega)$.

Meanwhile, the solutions $(u(\chi_k), v(\chi_k))$ to the state equation (1.23) and the adjoint equation (1.24) with χ replaced by χ_k are bounded in $[H_D^1(\Omega)^d]^2$, so $(u(\chi_k), v(\chi_k))$ weakly converges to some $(u^*, v^*) \in [H_D^1(\Omega)]^2$. According to [1], we have

$$(u^*, v^*) = \arg \min_{(u, v) \in [H_D^1(\Omega)]^2} \left(\frac{1}{2} a(\chi^*; u, u) - \ell_{in}(u) + \frac{1}{2} a(\chi^*; v, v) - \ell_{out}(v) \right).$$

Define $(\sigma^*, \varrho^*) := (\frac{1}{A(\chi^*)} E_0 \varepsilon(u^*), \frac{1}{A(\chi^*)} E_0 \varepsilon(v^*))$. In the following, we demonstrate that $\{(\chi^*, \sigma^*, \varrho^*)\}$ is a solution to the problem Q . Note that

$$\begin{aligned} & L^{\gamma,0}(\chi^*, \sigma^*, \varrho^*) - L^{\gamma,0}(\chi_k, \sigma(\chi_k), \varrho(\chi_k)) \\ &= \int_{\Omega} A(\chi^*) E_0^{-1} \sigma^* : \varrho^* \, dx + \frac{\gamma}{C_G} \|\chi^*\|_{TV} - \int_{\Omega} A(\chi_k) E_0^{-1} \sigma(\chi_k) : \varrho(\chi_k) \, dx - \frac{\gamma}{C_G} \|\chi_k\|_{TV} \\ &= \int_{\Gamma} t_{out} \cdot u^* \, dx + \frac{\gamma}{C_G} \|\chi^*\|_{TV} - \int_{\Gamma} t_{out} \cdot u(\chi_k) \, dx - \frac{\gamma}{C_G} \|\chi_k\|_{TV} \\ &= \int_{\Gamma} t_{out} \cdot (u^* - u(\chi_k)) \, ds + \frac{\gamma}{C_G} (\|\chi^*\|_{TV} - \|\chi_k\|_{TV}). \end{aligned}$$

Recall the classical trace inequality (cf. [4])

$$(2.4) \quad \|u\|_{0,\Gamma} \leq C \|u\|_{0,\Omega}^{1/2} \|u\|_{1,\Omega}^{1/2} \quad \forall u \in H^1(\Omega).$$

Since $u(\chi_k) \rightharpoonup u^*$ in $H^1(\Omega)^d$, it follows that $u(\chi_k) \rightarrow u^*$ in $L^2(\Omega)^d$. Thus, we have

$$\begin{aligned} & \int_{\Gamma} t_{out} \cdot (u^* - u(\chi_k)) \, ds \leq \|t_{out}\|_{0,\Gamma} \|u^* - u(\chi_k)\|_{0,\Gamma} \\ & \leq C_1 \|t_{out}\|_{0,\Gamma} \|u^* - u(\chi_k)\|_{0,\Omega}^{\frac{1}{2}} \|u^* - u(\chi_k)\|_{1,\Omega}^{\frac{1}{2}} \\ & \leq C_2 \|t_{out}\|_{0,\Gamma} (\|u^*\|_{1,\Omega} + \|u(\chi_k)\|_{1,\Omega})^{\frac{1}{2}} \|u^* - u(\chi_k)\|_{0,\Omega}^{\frac{1}{2}}. \end{aligned}$$

Furthermore, based on the lower semi-continuity of the TV-norm, we obtain

$$\begin{aligned} & \limsup_{h \downarrow 0} (L^{\gamma,0}(\chi^*, \sigma^*, \varrho^*) - L^{\gamma,0}(\chi_k, \sigma(\chi_k), \varrho(\chi_k))) \\ & \leq \limsup_{h \downarrow 0} C_2 \|t_{out}\|_{0,\Gamma} (\|u^*\|_{1,\Omega} + \|u(\chi_k)\|_{1,\Omega})^{\frac{1}{2}} \|u^* - u(\chi_k)\|_{0,\Omega}^{\frac{1}{2}} + \frac{\gamma}{C_G} \limsup_{h \downarrow 0} (\|\chi^*\|_{TV} - \|\chi_k\|_{TV}) \\ & \leq \frac{\gamma}{C_G} (\|\chi^*\|_{TV} - \liminf_{h \downarrow 0} \|\chi_k\|_{TV}) \leq 0. \end{aligned}$$

Therefore, we obtain the following inequality,

$$(2.5) \quad L^{\gamma,0}(\chi^*, \sigma^*, \varrho^*) \leq \liminf_{h \downarrow 0} L^{\gamma,0}(\chi_h, \sigma(\chi_h), \varrho(\chi_h)).$$

This shows that $(\chi^*, \sigma^*, \varrho^*)$ is a solution to the optimization problem Q . □

220 In the following we first investigate the convergence of the problem P to the problem Q .

221 **Theorem 2.11.** *For each $s > 2$, the functional L^{γ, ϵ_k} Γ -converges to $L^{\gamma, 0}$ in $X^s := \mathcal{U} \times (S \cap [L^s(\Omega)^{d \times d}]^2)$
222 with respect to the parameter ϵ_k , and L^{γ, ϵ_k} converges pointwise to $L^{\gamma, 0}$ in X .*

Proof. It follows from the definition that

$$L^{\gamma, \epsilon_k}(\chi, \sigma, \varrho) = L^{\epsilon_k}(\chi^{\epsilon_k}, \sigma, \varrho) + \frac{\gamma}{\epsilon_k} P_{G_{\epsilon_k}}(\chi),$$

$$L^{\gamma, 0}(\chi, \sigma, \varrho) = L^0(\chi, \sigma, \varrho) + \frac{\gamma}{C_G} \|\chi\|_{TV},$$

223 so we separately discuss the convergence of these two parts. By Lemma 2.7, we have established that
224 $\frac{\gamma}{\epsilon_k} P_{G_{\epsilon_k}}(\cdot)$ Γ -converges and pointwise converges to $\frac{\gamma}{C_G} \|\cdot\|_{TV}$. For $L^{\epsilon_k}(\cdot)$ and $L^0(\cdot)$, we will prove the
225 continuous convergence, denoted as $L^{\epsilon_k} \xrightarrow{c} L^0$.

226 Let $\chi_m \xrightarrow{L^1} \chi$ and $(\sigma_m, \varrho_m) \xrightarrow{L^s} (\sigma, \varrho)$ for some $s > 2$, it follows from the definition and (1.25) that

$$\begin{aligned} & L^{\epsilon_k}(\chi_m^{\epsilon_k}, \sigma_m, \varrho_m) - L^0(\chi, \sigma, \varrho) \\ &= \int_{\Omega} (A(\chi_m^{\epsilon_k}) - A(\chi)) E_0^{-1} \sigma_m : \varrho_m \, dx + \int_{\Omega} A(\chi) E_0^{-1} (\sigma_m - \sigma) : \varrho_m \, dx \\ & \quad + \int_{\Omega} A(\chi) E_0^{-1} \sigma : (\varrho_m - \varrho) \, dx + \lambda \int_{\Omega} (A(\chi_m^{\epsilon_k}) - A(\chi)) E_0^{-1} (|\sigma_m|^2 + |\varrho_m|^2) \, dx \\ & \quad + \lambda \int_{\Omega} A(\chi) E_0^{-1} (|\sigma_m|^2 - |\sigma|^2 + |\varrho_m|^2 - |\varrho|^2) \, dx + \lambda a(\chi; u(\chi), u(\chi)) \\ & \quad + \lambda a(\chi; v(\chi), v(\chi)) - \lambda a(\chi_m^{\epsilon_k}; u(\chi_m^{\epsilon_k}), u(\chi_m^{\epsilon_k})) - \lambda a(\chi_m^{\epsilon_k}; v(\chi_m^{\epsilon_k}), v(\chi_m^{\epsilon_k})). \end{aligned}$$

We first address the term involving $\chi_m^{\epsilon_k} - \chi$. Note that for elements in $\mathcal{U}$, L^1 -convergence is equivalent to L^p -convergence for any $p \in [1, +\infty)$. We take

$$\int_{\Omega} (A(\chi_m^{\epsilon_k}) - A(\chi)) E_0^{-1} \sigma_m : \varrho_m \, dx$$

227 as an example to illustrate the main idea of the proof. Recall the definition of $A(\chi)$, we have

$$\begin{aligned} & \int_{\Omega} (A(\chi_m^{\epsilon_k}) - A(\chi)) E_0^{-1} \sigma_m : \varrho_m \, dx \\ &= \int_{\Omega} (A(\chi_m^{\epsilon_k}) - A(\chi^{\epsilon_k})) E_0^{-1} \sigma_m : \varrho_m \, dx + \int_{\Omega} (A(\chi^{\epsilon_k}) - A(\chi)) E_0^{-1} \sigma_m : \varrho_m \, dx \\ &= \left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) \left(\int_{\Omega} G_{\epsilon_k} * (\chi_m - \chi) E_0^{-1} \sigma_m : \varrho_m \, dx + \int_{\Omega} (G_{\epsilon_k} * \chi - \chi) E_0^{-1} \sigma_m : \varrho_m \, dx \right) \\ &\leq \frac{C}{E_{\min}} \left(\|G_{\epsilon_k} * (\chi_m - \chi)\|_{0, \frac{s}{s-2}, \Omega} + \|G_{\epsilon_k} * \chi - \chi\|_{0, \frac{s}{s-2}, \Omega} \right) \|\sigma_m\|_{0, s, \Omega} \|\varrho_m\|_{0, s, \Omega} \\ &\leq \frac{C}{E_{\min}} \left(\|\chi_m - \chi\|_{0, \frac{s}{s-2}, \Omega} + \|G_{\epsilon_k} * \chi - \chi\|_{0, \frac{s}{s-2}, \Omega} \right) \|\sigma_m\|_{0, s, \Omega} \|\varrho_m\|_{0, s, \Omega} \rightarrow 0 \quad \text{as } k, m \rightarrow +\infty, \end{aligned}$$

228 where we used the generalized Hölder inequality in the first inequality and Young's inequality for con-
229 volutions (cf. Lemma 2.4) in the second inequality. The convergence of the first term in the limit is due
230 to $\chi_k \xrightarrow{L^p} \chi$, while the second term converges because of the convergence of the smooth Dirac sequence
231 (cf. Lemma 2.5).

232 The handling of $\sigma_m - \sigma$ and $\varrho_m - \varrho$ is straightforward. The key lies in the estimate of $u(\chi_m^{\epsilon_k}) - u(\chi)$,
233 which involves the stability of solutions to linear elasticity equations with respect to the elastic tensor.

$$\begin{aligned} a(\chi; u(\chi), u(\chi)) - a(\chi_m^{\epsilon_k}; u(\chi_m^{\epsilon_k}), u(\chi_m^{\epsilon_k})) &= \ell_{in}(u(\chi)) - \ell_{in}(u(\chi_m^{\epsilon_k})) \\ &= \|\ell_{in}\|_{0, \Gamma_N} \|u(\chi) - u(\chi_m^{\epsilon_k})\|_{0, \Gamma} \\ &\leq C \|\ell_{in}\|_{0, \Gamma_N} \|u(\chi) - u(\chi_m^{\epsilon_k})\|_{1, \Omega}. \end{aligned}$$

234 On the other hand, we have

$$a(\chi; u(\chi), \hat{u}) = \ell_{in}(\hat{u}) \quad \forall \hat{u} \in H_D^1(\Omega)^d, \quad a(\chi_m^{\epsilon_k}; u(\chi_m^{\epsilon_k}), \tilde{u}) = \ell_{in}(\tilde{u}) \quad \forall \tilde{u} \in H_D^1(\Omega)^d.$$

Setting $\hat{u} = \tilde{u} = u(\chi) - u(\chi_k^{\epsilon_k})$, we have

$$\begin{aligned} a(\chi; u(\chi) - u(\chi_m^{\epsilon_k}), u(\chi) - u(\chi_m^{\epsilon_k})) &= a(\chi_m^{\epsilon_k}; u(\chi_m^{\epsilon_k}), u(\chi) - u(\chi_m^{\epsilon_k})) - a(\chi; u(\chi_m^{\epsilon_k}), u(\chi) - u(\chi_m^{\epsilon_k})) \\ &= \left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) \int_{\Omega} (G_{\epsilon_k} * \chi_m - \chi) E_0 \varepsilon(u(\chi_m^{\epsilon_k})) : \varepsilon(u(\chi) - u(\chi_m^{\epsilon_k})) dx \\ &\leq \frac{C}{E_{\min}} \|G_{\epsilon_k} * \chi_m - \chi\|_{0, \frac{2s}{s-2}, \Omega} \|u(\chi_m^{\epsilon_k})\|_{1, s, \Omega} \|u(\chi) - u(\chi_m^{\epsilon_k})\|_{1, \Omega}. \end{aligned}$$

Hence,

$$\|u(\chi) - u(\chi_m^{\epsilon_k})\|_{1, \Omega} \leq C \frac{E_{\max}}{E_{\min}} \|G_{\epsilon_k} * \chi_m - \chi\|_{0, \frac{2s}{s-2}, \Omega} \|u(\chi_m^{\epsilon_k})\|_{1, s, \Omega}.$$

The remaining terms can also be handled sequentially, ultimately being controlled by $\|\chi_m - \chi\|_{0, p, \Omega}$, $\|G_{\epsilon_k} * \chi - \chi\|_{0, p, \Omega}$, $\|\sigma_m - \sigma\|_{0, s, \Omega}$, and $\|\varrho_m - \varrho\|_{0, s, \Omega}$. This completes the proof that $L^{\epsilon_k} \xrightarrow{c} L^0$. $\square$

Then we investigate the convergence of the problem P_n to the problem P .

Theorem 2.12. *For all $\epsilon_k > 0$, we have $L_n^{\gamma, \epsilon_k} \xrightarrow{sub-c} L^{\gamma, \epsilon_k}$ and $L_n^{\epsilon_k} \xrightarrow{sub-c} L^{\epsilon_k}$ as $n \rightarrow +\infty$. In particular, if the solutions to the linear elasticity equation have a higher regularity $(u, v) \in [H^s(\Omega)^d \cap H_D^1(\Omega)^d]^2$ for some $s > 1$, then for any sequence $\{\epsilon_n\}_{n \geq 0}$ with $\epsilon_n \rightarrow 0$, we have $L_n^{\epsilon_n} \xrightarrow{sub-c} L^0$.*

Proof. Let $\chi_n \xrightarrow{L^1} \chi$ and $(\sigma_n, \varrho_n) \xrightarrow{L^2} (\sigma, \varrho)$. We first examine the continuous convergence of the perimeter regularization. Since $\int_{\Omega} \chi_n dx \leq |\Omega| < \infty$, we have [9, eq. (1.2), Page 236]

$$(2.6) \quad P_{G_{\epsilon_k}}(\chi_n) = \int_{\Omega} \chi_n dx - \int_{\Omega} (G_{\epsilon_k} * \chi_n)(x) \chi_n(x) dx,$$

then

$$\begin{aligned} &|P_{G_{\epsilon_k}}(\chi_n) - P_{G_{\epsilon_k}}(\chi)| \\ &\leq \|\chi_n - \chi\|_{0, 1, \Omega} + \int_{\Omega} |(G_{\epsilon_k} * \chi_n) \chi_n - (G_{\epsilon_k} * \chi) \chi| dx \\ &\leq \|\chi_n - \chi\|_{0, 1, \Omega} + \int_{\Omega} |(G_{\epsilon_k} * \chi_n) \chi_n - (G_{\epsilon_k} * \chi_n) \chi| dx + \int_{\Omega} |(G_{\epsilon_k} * \chi_n) \chi - (G_{\epsilon_k} * \chi) \chi| dx \\ &\leq \|\chi_n - \chi\|_{0, 1, \Omega} + \int_{\Omega} (G_{\epsilon_k} * \chi_n) |\chi_n - \chi| dx + \int_{\Omega} |G_{\epsilon_k} * (\chi_n - \chi)| \chi dx \\ &\leq 3 \|\chi_n - \chi\|_{0, 1, \Omega}, \end{aligned}$$

where in the last inequality, we used Young's convolution inequality with $p = q = r = 1$ (cf. Lemma 2.4).

The following proof is very similar to that of Theorem 2.11, and even simpler, as we do not require $(\sigma, \varrho) \in [L^s(\Omega)^{d \times d}]^2$. Using again (1.25) and the definition, we have the following expression

$$\begin{aligned} &L_n^{\epsilon_k}(\chi_n^{\epsilon_k}, \sigma_n, \varrho_n) - L^{\epsilon_k}(\chi^{\epsilon_k}, \sigma, \varrho) \\ &= \int_{\Omega} (A(\chi_n^{\epsilon_k}) - A(\chi^{\epsilon_k})) E_0^{-1} \sigma_n : \varrho_n dx + \int_{\Omega} A(\chi^{\epsilon_k}) E_0^{-1} (\sigma_n - \sigma) : \varrho_n dx \\ &\quad + \int_{\Omega} A(\chi^{\epsilon_k}) E_0^{-1} \sigma : (\varrho_n - \varrho) dx + \lambda \int_{\Omega} (A(\chi_n^{\epsilon_k}) - A(\chi^{\epsilon_k})) E_0^{-1} (|\sigma_n|^2 + |\varrho_n|^2) dx \\ &\quad + \lambda \int_{\Omega} A(\chi^{\epsilon_k}) E_0^{-1} (|\sigma_n|^2 - |\sigma|^2 + |\varrho_n|^2 - |\varrho|^2) dx + \lambda a(\chi^{\epsilon_k}; u(\chi^{\epsilon_k}), u(\chi^{\epsilon_k})) \\ &\quad + \lambda a(\chi^{\epsilon_k}; v(\chi^{\epsilon_k}), v(\chi^{\epsilon_k})) - \lambda a(\chi_n^{\epsilon_k}; u_n(\chi_n^{\epsilon_k}), u_n(\chi_n^{\epsilon_k})) - \lambda a(\chi_n^{\epsilon_k}; v_n(\chi_n^{\epsilon_k}), v_n(\chi_n^{\epsilon_k})), \end{aligned}$$

which can be easily decomposed into terms involving $\chi_n - \chi$, $\sigma_n - \sigma$, $\varrho_n - \varrho$, $u_n - u(\chi)$, and $v_n - v(\chi)$. The estimates for terms containing $\sigma_n - \sigma$ and $\varrho_n - \varrho$ are straightforward.

We first estimate the terms containing $\chi_n - \chi$ as follows

$$\begin{aligned} \int_{\Omega} (A(\chi_n^{\epsilon_k}) - A(\chi^{\epsilon_k})) E_0^{-1} \sigma_n : \varrho_n dx &= \left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) \int_{\Omega} G_{\epsilon_k} * (\chi_n - \chi) E_0^{-1} \sigma_n : \varrho_n dx \\ &\leq \frac{C}{E_{\min}} \|G_{\epsilon_k} * (\chi_n - \chi)\|_{0, \infty, \Omega} \|\sigma_n\|_{0, \Omega} \|\varrho_n\|_{0, \Omega} \\ &\leq \frac{C}{E_{\min}} \|G_{\epsilon_k}\|_{0, \Omega} \|\chi_n - \chi\|_{0, \Omega} \|\sigma_n\|_{0, \Omega} \|\varrho_n\|_{0, \Omega} \rightarrow 0 \quad \text{as } n \rightarrow +\infty, \end{aligned}$$

where we employed Young's convolution inequality in the last inequality. The boundedness of $\|G_{\epsilon_k}\|_{0,\Omega}$ is guaranteed because ϵ_k is fixed at this point.

Then we estimate the terms containing $u_n(\chi_n^{\epsilon_k}) - u(\chi^{\epsilon_k})$ below

$$\begin{aligned} & a(\chi^{\epsilon_k}; u(\chi^{\epsilon_k}), u(\chi^{\epsilon_k})) - a(\chi_n^{\epsilon_k}; u_n(\chi_n^{\epsilon_k}), u_n(\chi_n^{\epsilon_k})) \\ &= \ell_{in}(u(\chi^{\epsilon_k})) - \ell_{in}(u_n(\chi_n^{\epsilon_k})) \\ &= \ell_{in}(u(\chi^{\epsilon_k}) - u_n(\chi^{\epsilon_k})) + \ell_{in}(u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})) \\ &\leq \|\ell_{in}\|_{0,\Gamma_N} \|u(\chi^{\epsilon_k}) - u_n(\chi^{\epsilon_k})\|_{0,\Gamma_N} + \|\ell_{in}\|_{0,\Gamma_N} \|u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})\|_{0,\Gamma_N} \\ &\leq C \|\ell_{in}\|_{0,\Gamma_N} [\|u(\chi^{\epsilon_k}) - u_n(\chi^{\epsilon_k})\|_{1,\Omega} + \|u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})\|_{1,\Omega}]. \end{aligned}$$

The convergence of the first term can be obtained by using classical finite element error estimates

$$\|u(\chi^{\epsilon_k}) - u_n(\chi^{\epsilon_k})\|_{1,\Omega} \leq Ch_n^{s-1} \|u(\chi^{\epsilon_k})\|_{s,\Omega} \leq Ch_n^{s-1} \|\ell_{in}\|_{0,\Gamma_N}.$$

On the other hand, we have

$$a(\chi^{\epsilon_k}; u_n(\chi^{\epsilon_k}), \hat{u}_n) = \ell_{in}(\hat{u}_n) \quad \forall \hat{u}_n \in V_n, \quad a(\chi_n^{\epsilon_k}; u_n(\chi_n^{\epsilon_k}), \tilde{u}_n) = \ell_{in}(\tilde{u}_n) \quad \forall \tilde{u}_n \in V_n,$$

where $V_n \subset H_D^1(\Omega)^d$ is finite element space.

Setting $\hat{u}_n = \tilde{u}_n = u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})$ we obtain

$$\begin{aligned} & a(\chi^{\epsilon_k}; u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k}), u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})) \\ &= a(\chi_n^{\epsilon_k}; u_n(\chi_n^{\epsilon_k}), u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})) - a(\chi^{\epsilon_k}; u_n(\chi_n^{\epsilon_k}), u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})) \\ &= \left(\frac{1}{E_{\max}} - \frac{1}{E_{\min}} \right) \int_{\Omega} G_{\epsilon_k} * (\chi_n - \chi) E_0 \varepsilon(u_n(\chi_n^{\epsilon_k})) : \varepsilon(u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})) dx \\ &\leq \frac{C}{E_{\min}} \|G_{\epsilon_k} * (\chi_n - \chi)\|_{0,\infty,\Omega} \|u_n(\chi_n^{\epsilon_k})\|_{1,\Omega} \|u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})\|_{1,\Omega} \\ &\leq \frac{C}{E_{\min}} \|G_{\epsilon_k}\|_{0,\Omega} \|\chi_n - \chi\|_{0,\Omega} \|u_n(\chi_n^{\epsilon_k})\|_{1,\Omega} \|u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})\|_{1,\Omega}. \end{aligned}$$

Hence,

$$\|u_n(\chi^{\epsilon_k}) - u_n(\chi_n^{\epsilon_k})\|_{1,\Omega} \leq C \frac{E_{\max}}{E_{\min}} \|G_{\epsilon_k}\|_{0,\Omega} \|\chi_n - \chi\|_{0,\Omega} \|u_n(\chi_n^{\epsilon_k})\|_{1,\Omega}.$$

The remaining terms can also be handled sequentially.

It is easy to see that among all the estimates above, only the finite element error estimate depends on the value of ϵ_k . When $u \in H^s(\Omega)^d \cap H_D^1(\Omega)^d$, the finite element error estimate has a convergence order of $s - 1$, depending only on the mesh size h_n . At this point, the estimates will ultimately be controlled by $\|\chi_n - \chi\|_{0,\Omega}$, $\|\sigma_n - \sigma\|_{0,\Omega}$, and $\|\varrho_n - \varrho\|_{0,\Omega}$, we obtain the following

$$L_n^{\gamma,\epsilon_k} \xrightarrow{sub-c} L^{\gamma,\epsilon_k}, \quad L_n^{\epsilon_k} \xrightarrow{sub-c} L^{\epsilon_k} \quad \text{for any } \epsilon_k \geq 0 \text{ as } n \rightarrow +\infty.$$

If $u \in H^s(\Omega)^d \cap H_D^1(\Omega)^d$, then

$$L_n^{\epsilon_n} \xrightarrow{sub-c} L^0 \quad \text{as } n \rightarrow +\infty.$$

This finishes the proof. $\square$

We remark that the requirement $u \in H^s(\Omega)^d \cap H_D^1(\Omega)^d$ in Theorem 2.12 is not indispensable. In fact, we can show the strong convergence of $u_n(\chi^{\epsilon_k})$ to $u(\chi^{\epsilon_k})$ as $h_n \rightarrow 0$ by using density arguments.

Now we are ready to prove the convergence of the problem P_n to problem Q .

Theorem 2.13. *For each $s > 2$ and $N > 0$, assume the sequence $\{X_n\}$ is nested such that $X_1 \subset X_2 \subset \dots \subset X_N$. Then, for any sequence $\{\epsilon_n\}_{n \geq 1}$, with $\epsilon_n \rightarrow 0$, we have the functional L_n^{γ,ϵ_n} Γ -converges to $L^{\gamma,0}$ in X^s with respect to parameters n , while L_n^{γ,ϵ_n} converges pointwise to $L^{\gamma,0}$ in $\bigcup_{n \leq N} X_n$.*

Proof. We first note that the domains of L_n^{γ,ϵ_k} and $L^{\gamma,0}$ are different. However, since we use the conforming finite element method, we have $X_n \subset X$. Thus, we can naturally extend the definition of L_n to X , and then discuss Γ -convergence and pointwise convergence without ambiguity.

We consider a larger family of functions, where the subscripts and superscripts are denoted by ϵ_k and n , respectively. For any ϵ_k and n , we extend the definition as follows:

$$(2.7) \quad \bar{L}_n^{\gamma,\epsilon_k}(\chi, \sigma, \varrho) = \begin{cases} L_n^{\gamma,\epsilon_k}(\chi, \sigma, \varrho) & \text{if } (\chi, \sigma, \varrho) \in X_n, \\ +\infty & \text{if } (\chi, \sigma, \varrho) \notin X_n. \end{cases}$$

Next, we proceed to prove Γ -convergence. Given $(\chi, \sigma, \varrho) \in X^s$, we have the following conclusions

(1) For all $(\chi_n, \sigma_n, \varrho_n) \rightarrow (\chi, \sigma, \varrho)$ in the space $(L^1(\Omega), L^s(\Omega), L^s(\Omega))$, we have

$$\begin{aligned} & \bar{L}_n^{\gamma, \epsilon_n}(\chi_n, \sigma_n, \varrho_n) - L^{\gamma, 0}(\chi, \sigma, \varrho) \\ &= [\bar{L}_n^{\epsilon_n}(\chi_n, \sigma_n, \varrho_n) - L^0(\chi, \sigma, \varrho)] + \left[\frac{\gamma}{\epsilon_n} P_{G_{\epsilon_n}}(\chi_n) - \frac{\gamma}{C_G} \|\chi\|_{TV} \right]. \end{aligned}$$

As $n \rightarrow +\infty$, the first term is known to be greater than or converges to 0 by Theorem 2.12 (sub-continuous convergence of $L_n^{\epsilon_n}$), while the second term is greater than 0 by Lemma 2.7 (lower semicontinuity by the Γ -convergence of the nonlocal perimeter approximation). Therefore, we ultimately have

$$\liminf_{n \rightarrow +\infty} \bar{L}_n^{\gamma, \epsilon_n}(\chi_n, \sigma_n, \varrho_n) \geq L^{\gamma, 0}(\chi, \sigma, \varrho).$$

(2) With the Γ -convergence of L^{γ, ϵ_k} to $L^{\gamma, 0}$ established in Theorem 2.11, we know that there exists a sequence $\{(\chi_k, \sigma_k, \varrho_k)\}_{k \geq 1} \subset X^s$ such that

$$\limsup_{k \rightarrow \infty} L^{\gamma, \epsilon_k}(\chi_k, \sigma_k, \varrho_k) \leq L^{\gamma, 0}(\chi, \sigma, \varrho).$$

By the definition of sub-continuous convergence, for each k and any $\{(\chi_{k,n}, \sigma_{k,n}, \varrho_{k,n})\} \subset X_n \subset X$, we have

$$\lim_{n \rightarrow \infty} L_n^{\gamma, \epsilon_k}(\chi_{k,n}, \sigma_{k,n}, \varrho_{k,n}) = L^{\gamma, \epsilon_k}(\chi_k, \sigma_k, \varrho_k).$$

Thus, we take the recovery sequence $\{(\chi_{k,k}, \sigma_{k,k}, \varrho_{k,k})\}$ and obtain

$$\limsup_{k \rightarrow \infty} \bar{L}_k^{\gamma, \epsilon_k}(\chi_{k,k}, \sigma_{k,k}, \varrho_{k,k}) = \limsup_{k \rightarrow \infty} L_k^{\gamma, \epsilon_k}(\chi_{k,k}, \sigma_{k,k}, \varrho_{k,k}) \leq L^{\gamma, 0}(\chi, \sigma, \varrho).$$

Recall Definition 2.1, combining (1) and (2) we prove $\bar{L}_n^{\gamma, \epsilon_n} \xrightarrow{\Gamma} L^{\gamma, 0}$ in X^s .

In the following, we consider pointwise convergence. For any $(\chi_N, \sigma_N, \varrho_N) \in \bigcup_{n' \leq N} X_{n'}$, as long as $n > N$, we have $(\chi_N, \sigma_N, \varrho_N) \in X_n$. At this point, $\bar{L}_n^{\gamma, \epsilon_n}(\chi_N, \sigma_N, \varrho_N) = L_n^{\gamma, \epsilon_n}(\chi_N, \sigma_N, \varrho_N)$. Therefore, it suffices to use the pointwise convergence established in Lemma 2.7 and Theorem 2.12 to conclude

$$\lim_{n \rightarrow \infty} \bar{L}_n^{\gamma, \epsilon_n}(\chi_N, \sigma_N, \varrho_N) = \lim_{n \rightarrow \infty} \left(L_n^{\epsilon_n}(\chi_N^{\epsilon_n}, \sigma_N, \varrho_N) + \frac{\gamma}{C_G} P_{G_{\epsilon_n}}(\chi_N) \right) = L^{\gamma, 0}(\chi_N, \sigma_N, \varrho_N).$$

This finishes the proof. $\square$

We note that in Theorem 2.13, we actually proved the convergence of $\bar{L}_n^{\gamma, \epsilon_k}$ to $L^{\gamma, 0}$. The reason we view this as the convergence of L_n^{γ, ϵ_k} to $L^{\gamma, 0}$ is that, for our research subject—the convergence between optimization problems—the minimizers of $\bar{L}_n^{\gamma, \epsilon_k}$ and L_n^{γ, ϵ_k} are completely equivalent. Henceforth, we no longer distinguish between $\bar{L}_n^{\gamma, \epsilon_k}$ and L_n^{γ, ϵ_k} , and directly treat L_n^{γ, ϵ_k} as defined on X .

We can illustrate the relationships among the three problems Q , P and P_n using the following commuting diagram:

$$\begin{array}{ccc} L^{\gamma, \epsilon_k} & \xrightarrow{k} & L^{\gamma, 0} \\ \uparrow n & \nearrow n, k & \\ L_n^{\gamma, \epsilon_k} & & \end{array}$$

2.4. Convergence characterization. From the above commuting diagram, we can observe two paths for the convergence of problem P_n to problem Q , where ϵ_k and h_n are independent in the limit process. However, when considering the convergence of local minimizers, we can no longer ignore this issue, as the existence of a solution to problem P is not yet guaranteed. Therefore, when solving problem Q , we cannot first refine the mesh and then adjust the perimeter approximation parameter ϵ_k , as this cannot ensure the convergence of minimizers. We first prove the following convergence theorem for local minimizers.

Theorem 2.14. *Let $\{\chi_n\}$ be a sequence of local minimizers of the problem $\{P_n\}$ with an appropriate chosen ϵ_n for each n . For every accumulation point χ^+ of $\{\chi_n\}$, if $\chi^+ \in U$ with U the minimal neighborhood (in the L^1 -metric) of some isolated local minimizer χ^* of problem Q , then $\chi^+ = \chi^*$.*

Proof. It suffices to verify the conditions of Lemma 2.9. We have already proved that L_n^{γ, ϵ_n} Γ -converges to $L^{\gamma, 0}$, while the coercivity of L_n^{γ, ϵ_n} (in the L^1 -metric) and its lower semicontinuity are evident. The key step now is to verify the equicoercivity.

311 We first prove that $L_n^{\epsilon_n}$ has a uniform lower bound M . In Section 1, we have shown that $L_n^{\epsilon_n}(\chi_n^{\epsilon_n}, \sigma_n^*, \varrho_n^*) =$
 312 $\min_{(\hat{\sigma}_n, \hat{\varrho}_n) \in S_n} L_n^{\epsilon_n}(\chi_n^{\epsilon_n}, \hat{\sigma}_n, \hat{\varrho}_n)$. Thus,

$$\begin{aligned} L_n^{\epsilon_n}(\chi_n^{\epsilon_n}, \hat{\sigma}_n, \hat{\varrho}_n) &\geq L_n^{\epsilon_n}(\chi_n^{\epsilon_n}, \sigma_n^*, \varrho_n^*) \\ &= a(\chi_n^{\epsilon_n}; u_n^*, v_n^*) \\ &= \ell_{out}(u_n^*) \\ &\geq -C_1 \|\ell_{out}\|_{0, \Gamma_N} \|u_n^*\|_{1, \Omega} \\ &\geq -C_2 \|\ell_{out}\|_{0, \Gamma_N} \|\ell_{in}\|_{0, \Gamma_N}. \end{aligned}$$

313 Therefore, $M := -C_2 \|\ell_{out}\|_{0, \Gamma_N} \|\ell_{in}\|_{0, \Gamma_N} \leq L_n^{\epsilon_n}(\chi_n^{\epsilon_n}, \sigma_n, \varrho_n)$ for all n, ϵ_n and $(\chi_n, \sigma_n, \varrho_n) \in X$.
 Let $t \in \mathbb{R}$, we choose the set

$$K_t = \left\{ \chi \in \mathcal{U} : \frac{\gamma}{C_G} \|\chi\|_{TV} \leq \max(t - M + 1, 1) \right\}.$$

Since $BV(\Omega)$ is compactly embedded in $L^1(\Omega)$ and K_t is bounded in $BV(\Omega)$, it is a compact set in $L^1(\Omega)$. For any n , we note that $\mathcal{U}_n$ is a finite set and Lemma 2.9 have

$$\lim_{\epsilon \downarrow 0} \frac{\gamma}{\epsilon} P_{G_\epsilon}(\chi_{n,i}) = \frac{\gamma}{C_G} \|\chi_{n,i}\|_{TV} \quad \forall \chi_{n,i} \in \mathcal{U}_n.$$

Hence for each $\delta > 0$, fix $\chi_{n,i} \in \mathcal{U}_n$, there exists a sufficiently small $\epsilon'_{n,i} > 0$, for each $\epsilon_{n,i} \in (0, \epsilon'_{n,i})$ have

$$\left| \frac{\gamma}{\epsilon_{n,i}} P_{G_{\epsilon_{n,i}}}(\chi_{n,i}) - \frac{\gamma}{C_G} \|\chi_{n,i}\|_{TV} \right| < \delta.$$

We set $\delta = 1$, $\epsilon_n = \min_i \epsilon_{n,i}$, if $(\chi_n, \sigma_n, \rho_n) \in X_n$ such that

$$L_n^{\gamma, \epsilon_n}(\chi_n, \sigma_n, \rho_n) \leq t,$$

then

$$\frac{\gamma}{C_G} \|\chi_n\|_{TV} \leq \frac{\gamma}{\epsilon_n} P_{G_{\epsilon_n}}(\chi_n) + 1 \leq t - L_n^{\epsilon_n}(\chi_n^{\epsilon_n}, \sigma_n, \rho_n) + 1 \leq t - M + 1.$$

314 Hence, $\{\chi_n : L_n^{\gamma, \epsilon_n}(\chi_n, \sigma_n, \rho_n) < t\} \subset K_t$ for any $n \geq 1$, i.e., the sequence $\{L_n^{\gamma, \epsilon_n}\}$ satisfies the equicoer-
 315 civity. We thus obtain the conclusion of the theorem. $\square$

316 **Theorem 2.15.** *Algorithm 2 terminates finitely at a local minimizer of the discrete problem.*

317 *Proof.* To clarify the notation again: $\mathcal{U}_n$ denotes the feasible set of the variable χ_n in the discrete problem
 318 P_n , $\chi_{n,k}$ represents the k -th iteration of the discrete problem corresponding to n $\chi_{n,k}^{\epsilon_n} = G_{\epsilon_n} * \chi_{h,k}$.

319 The finite termination of Algorithm 2 is evident, as the discrete problem is essentially a 0-1 pro-
 320 gramming problem, and Algorithm 2 ensures a strict monotonic descent of the objective function. This
 321 implies that the variable $\chi_{n,k}$ at each iteration step k will not repeat, and since the feasible set is finite,
 322 the algorithm iterates at most a finite number of steps. Therefore, to demonstrate that the algorithm
 323 must terminate finitely, it suffices to show that the line search also executes at most a finite number of
 324 steps.

325 In fact, recalling the update method of the discrete problem, the line search guarantees the following
 326 fact: when $r = r_{\min}$, $\chi_{h,k}$ is not updated; when $r = r_{\max}$, there are only three cases:

- 327 • The update of $\chi_{h,k}$ does not exceed the minimum update amount, but the objective function
 328 increases, the line search terminates.
- 329 • The value of the updated function decreases and the line search terminates.
- 330 • The value of the updated function does not decrease, and the update of $\chi_{h,k}$ exceeds the minimum
 331 update amount, leading to a contraction of the step size.

332 Since the update in the discrete problem requires projection onto the discrete feasible set, for the
 333 variable $\chi_{h,k}$ at the k -th step, there must exist an open neighborhood in $\text{conv}(\mathcal{U}_h)$ (under the L^1 met-
 334 ric) such that all elements in the neighborhood are projected to $\chi_{h,k}$. In the following, we rigorously
 335 demonstrate this fact: From $\chi_k = (1 - r\gamma)\chi_{h,k} - r(ab - \gamma\chi_{h,k}^\epsilon)$, we have

$$(2.8) \quad \chi_k = \chi_{h,k} - r(ab + \gamma(\chi_{h,k} - \chi_{h,k}^\epsilon)),$$

This is essentially a gradient descent method, notation $g = ab + \gamma(\chi_{h,k} - \chi_{h,k}^\epsilon)$. As the iterations progress,
 the search interval for the step size r becomes increasingly smaller ($r < \frac{1}{2\|g\|_\infty}$), while always satisfying
 that $r_{\min}$ does not update $\chi_{h,k}$. Because of

$$(\chi_{h,k})_i - rg_i > 1 - \frac{1}{2} = \frac{1}{2} \quad \text{if} \quad (\chi_{h,k})_i = 1,$$

$$(\chi_{h,k})_i - rg_i < 0 + \frac{1}{2} = \frac{1}{2} \quad \text{if} \quad (\chi_{h,k})_i = 0,$$

hence $P_{\mathcal{U}_h}(\chi_k) = \chi_{h,k}$.

Thus, if the line search can proceed for a certain number of steps, when $r_{\max} - r_{\min} < \frac{1}{2\|g\|_\infty}$, the update amount does not exceed two components, so at this point, the line search will terminate regardless of whether the objective function decreases, $g_{\max}$ and $g_{\max 2}$ are the absolute value of the first and second largest component of the vector $ab + \gamma(\chi_{h,k} - \chi_{h,k}^\epsilon)$ in terms of absolute value, respectively.

we prove that as long as $\chi_{h,k}$ is not a global minimizer, it must be possible to do so. **continue...**

□

3. ESTABLISH AN ABSTRACT THEORETICAL FRAMEWORK

4. THRESHOLD DYNAMICS METHOD FOR THE HEAT CONDUCTION EQUATION

In the work of Chen et al.[6], their predictor-corrector threshold dynamics method first obtains a search direction through threshold-driven operations, then monotonically decreases the objective function via step-size contraction along this direction. However, this scheme exhibits two critical limitations:

- (i) Failure to guarantee the existence of an objective-decreasing step size along the proposed direction;
- (ii) Each search step requires solving a PDE, which incurs substantial computational costs.

Our improved methodology eliminates line search requirements while maintaining the computational efficiency of the threshold dynamics framework. However, two unresolved aspects persist: (i) incomplete characterization of parameter estimation bounds and (ii) lack of theoretical analysis regarding its sensitivity to initial domain configurations.

We consider the following topology optimization of heat transfer problems.

$$(4.1) \quad \min_{(\chi, T) \in \mathcal{U} \times H_D^1(\Omega)} J(\chi, T) = \int_{\Omega} q(\chi) T dx$$

subject to

$$(4.2) \quad \int_{\Omega} k(\chi) \nabla T \cdot \nabla S dx = \int_{\Omega} q(\chi) S dx \quad \forall S \in H_D^1(\Omega),$$

where χ is an indicator function of the shape. We define the admissible set of feasible shapes

$$\mathcal{U} = \{\chi \in BV(\Omega) : \chi \in \{0, 1\}, \int_{\Omega} \chi dx \leq \beta |\Omega|\},$$

the heat source

$$q(\chi) = q_1 \chi + q_2(1 - \chi), \quad (q_1 < q_2),$$

as well as the heat conductivity coefficient

$$k(\chi) = k_1 \chi + k_2(1 - \chi), \quad (k_1 > k_2).$$

We impose a homogeneous Dirichlet boundary condition, while inhomogeneous problems can be transformed into homogeneous ones by modifying the source term.

We introduce a new variable σ , defined as

$$(4.3) \quad \sigma = k(\chi) \nabla T,$$

This allows the above optimization problem to be transformed into the following one.

$$(4.4) \quad \min_{\chi, \sigma} \hat{J}(\chi, \sigma) = \int_{\Omega} k(\chi)^{-1} \sigma \cdot \sigma dx$$

subject to

$$(4.5) \quad \sigma \in S(\chi) := \{\sigma \in L^2(\Omega)^d : \nabla \cdot \sigma = q(\chi), \sigma \cdot \vec{n} = 0\}.$$

Specifically, the modified objective functional is introduced as follows (since the space of σ depends on χ , a relaxation λ is introduced):

$$(4.6) \quad L(\chi, \sigma, \lambda) = \int_{\Omega} k(\chi)^{-1} \sigma \cdot \sigma dx + \frac{\alpha}{2} \left(\int_{\Omega} |\lambda - q(\chi)|^2 \right)^{\frac{1}{2}},$$

where

$$(4.7) \quad (\sigma, \lambda) \in SL := \{(\sigma, \lambda) \in [L^2(\Omega)^d]^2 : \nabla \cdot \sigma = \lambda, \sigma \cdot \vec{n}|_{\Gamma_N} = 0\}.$$

We can easily prove that, provided that α is large enough, problems (1.1)-(1.2) and (1.6)-(1.7) are equivalent.

366 **Lemma 4.1.** *If $\alpha \leq Cq_2|\Omega|$, then*

$$(4.8) \quad J(\chi, T^*) = \min_{(\sigma, \lambda)} L(\chi, \sigma, \lambda),$$

367 *where T^* is the unique solution to the variational problem (1.2).*

Proof. Since L is a convex, lower semi-continuous function with respect to (σ, λ) , and bounded from below, and SL is a convex set, L has a unique minimizer in SL . Thus, it suffices to prove that

$$(\sigma^*, \lambda^*) := (k(\chi)\nabla T^*, q(\chi))$$

368 is the minimizer. For each $(\sigma, \lambda) \in SL$, we have

$$\begin{aligned} & L(\chi, \sigma, \lambda) - L(\chi, \sigma^*, q(\chi)) \\ &= \int_{\Omega} k(\chi)^{-1}(\sigma \cdot \sigma - \sigma^* \cdot \sigma^*)dx + \frac{\alpha}{2}\|\lambda - q(\chi)\|_{0,\Omega} \\ &= \int_{\Omega} k(\chi)^{-1}|\sigma - \sigma^*|^2dx + 2 \int_{\Omega} k(\chi)^{-1}(\sigma - \sigma^*) \cdot \sigma^*dx + \frac{\alpha}{2}\|\lambda - q(\chi)\|_{0,\Omega} \\ &= \int_{\Omega} k(\chi)^{-1}|\sigma - \sigma^*|^2dx + 2 \int_{\Omega} \nabla T^* \cdot (\sigma - \sigma^*)dx + \frac{\alpha}{2}\|\lambda - q(\chi)\|_{0,\Omega} \\ &= \int_{\Omega} k(\chi)^{-1}|\sigma - \sigma^*|^2dx + \frac{\alpha}{2}\|\lambda - q(\chi)\|_{0,\Omega} - 2 \int_{\Omega} T^* \nabla \cdot (\sigma - \sigma^*)dx \\ &\geq \int_{\Omega} k(\chi)^{-1}|\sigma - \sigma^*|^2dx + \frac{\alpha}{2}\|\lambda - q(\chi)\|_{0,\Omega} - 2\|T^*\|_{0,\Omega}\|\lambda - q(\chi)\|_{0,\Omega} \\ &= \int_{\Omega} k(\chi)^{-1}|\sigma - \sigma^*|^2dx + \left(\frac{\alpha}{2} - 2\|T^*\|_{0,\Omega}\right)\|\lambda - q(\chi)\|_{0,\Omega}. \end{aligned}$$

369 On the other hand, based on the regularity of elliptic equations and Poincare's inequality, we have
370 the following inequality:

$$(4.9) \quad \|T^*\|_{0,\Omega} \leq C_1|T^*|_{1,\Omega} \leq C_2\|q(\chi)\|_{0,\Omega} \leq C_2q_2|\Omega|.$$

371 This finishes the proof. $\square$

372 The lower bound for α is overestimated, and numerical experiments reveal that the lower bound of α
373 is actually much smaller.

374 Next, a convolution approximation is applied to problem (1.6)-(1.7), and a perimeter constraint is
375 added to ensure that the optimization problem is well-posed.

376 Denote

$$\begin{aligned} \chi^\tau &= G_\tau * \chi, \quad q^\tau = q(\chi^\tau), \\ k^\tau &= G_\tau * k(\chi), \quad k^{-\tau} = G_\tau * (k(\chi)^{-1}), \end{aligned}$$

377 we have

$$(4.10) \quad \min_{\chi} \min_{(\sigma, \lambda)} L^\tau(\chi, \sigma, \lambda) = \int_{\Omega} k^{-\tau} \sigma \cdot \sigma dx + \frac{\alpha}{2}\|\lambda - q^\tau\|_{0,\Omega} + \gamma \int_{\Omega} (1 - \chi) \chi^\tau dx.$$

378 Consequently, the problem becomes a double minimization problem, where the inner minimization prob-
379 lem can be solved by addressing the corresponding heat equation, namely by solving the following
380 problem:

$$(4.11) \quad \int_{\Omega} (k^{-\tau})^{-1} \nabla T \cdot \nabla S dx = \int_{\Omega} q^\tau S dx \quad \forall S \in H_D^1(\Omega).$$

381 It should be noted that after applying the Gaussian heat kernel convolution, $k^{-\tau} \neq (k^\tau)^{-1}$.

382 In the following we discuss how to solve the outer minimization problem, ensuring that the objective
383 functional exhibits monotonic descent property.

384 At the discrete level, it is straightforward to construct a monotonically decreasing scheme, but this
385 imposes requirements on the parameter α . Although numerical experiments show good performance,
386 theoretically satisfactory results have not been achieved. In the following, we introduce a more effective
387 objective function.

$$(4.12) \quad L(\chi, \sigma, \lambda) = \int_{\Omega} k^{-1} \sigma \cdot \sigma dx + \alpha \int_{\Omega} (2\chi - 1)(\lambda - q_2) dx + (q_2 - q_1)|\Omega_1|,$$

388 where

$$(4.13) \quad (\sigma, \lambda) \in \mathcal{S} := \{(\sigma, \lambda) \in L^2(\Omega)^2 : -\nabla \cdot \sigma = \lambda, \sigma \cdot \vec{n}|_{\Gamma_N}, \lambda(x) \in [q_1, q_2]\}.$$

Lemma 4.2. *If α is big enough, then*

$$(4.14) \quad J(\chi, T^*) = \min_{(\sigma, \lambda)} L(\chi, \sigma, \lambda),$$

where T^* is the unique solution to the variational problem (1.2).

Proof.

$$\begin{aligned} & L(\chi, \sigma, \lambda) - L(\chi, \sigma^*, q) \\ &= \int_{\Omega} k^{-1} \sigma \cdot \sigma dx + \alpha \int_{\Omega} (2\chi - 1)(\lambda - q_2) dx \\ & \quad + \alpha \int_{\Omega} (q_2 - q_1) \chi dx - \int_{\Omega} k^{-1} \sigma^* \cdot \sigma^* dx \\ &= \int_{\Omega} k^{-1} |\sigma - \sigma^*|^2 dx + 2 \int_{\Omega} T(\lambda - q) dx + \alpha \int_{\Omega} (q_2 - q_1) \chi dx \\ & \quad + \alpha \int_{\Omega} (2\chi - 1)(\lambda - q_2) dx. \end{aligned}$$

Notice,

$$\begin{aligned} 2 \int_{\Omega} T(\lambda - q) dx &= 2 \int_{\Omega} T(\lambda - q_1) \chi dx + 2 \int_{\Omega} T(\lambda - q_2)(1 - \chi) dx, \\ \alpha \int_{\Omega} (2\chi - 1)(\lambda - q_2) dx &= \alpha \int_{\Omega} \chi(\lambda - q_2) dx + \alpha \int_{\Omega} (1 - \chi)(q_2 - \lambda) dx. \end{aligned}$$

Hence,

$$\begin{aligned} & 2 \int_{\Omega} T(\lambda - q) dx + \alpha \int_{\Omega} (q_2 - q_1) \chi dx + \alpha \int_{\Omega} (2\chi - 1)(\lambda - q_2) dx \\ &= \alpha \int_{\Omega} \chi(\lambda - q_1) dx + \int_{\Omega} (1 - \chi)(q_2 - \lambda)(\alpha - 2T) dx + 2 \int_{\Omega} T(\lambda - q_1) \chi dx \\ &= \int_{\Omega} (\alpha + 2T) \chi(\lambda - q_1) dx + \int_{\Omega} (\alpha - 2T)(1 - \chi)(q_2 - \lambda) dx. \end{aligned}$$

This finishes the proof. $\square$

It is readily observed that L is linear with respect to χ , and thus the function L^ε , with the perimeter constraint, is strictly concave. Consequently, the monotonic descent property of the algorithm can be easily derived.

Algorithm 3 Monotonic Descent Algorithm for shape optimization of heat equation

- 1: Initialize χ (The initial state is typically set to a uniform distribution with $\chi = \beta$), and parameter α, γ, τ
 - 2: **while** χ not converged **do**
 - 3: Solve $\int_{\Omega} (k^{-\tau})^{-1} \nabla T \cdot \nabla S dx = \int_{\Omega} q^\tau S dx, \quad \forall S \in H_D^1(\Omega)$
 - 4: Compute $\Phi^k(x) = \left(\frac{1}{k_1} - \frac{1}{k_2} \right) (\sigma^k \cdot \sigma^k)^\tau + 2\alpha(q_1 - q_2)(\chi^k)^\tau + \gamma \sqrt{\frac{\pi}{\tau}} (1 - 2\chi)^\tau$
 - 5: Update $\chi^{k+1} \leftarrow \chi^k$ to satisfy $\chi^{k+1} \in \mathcal{U}, \int_{\Omega} \chi^{k+1} \Phi^k dx \leq \int_{\Omega} \chi \Phi^k dx \quad \forall \chi \in \mathcal{U}$
 - 6: **end while**
 - 7: **return** Optimal shape χ^* and function value $L(\chi^*, \sigma^*, \lambda^*)$
-

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SKLMS, INSTITUTE OF COMPUTATIONAL MATHEMATICS AND SCIENTIFIC/ENGINEERING COMPUTING, ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING, CHINA.
Email address: wgong@lsec.cc.ac.cn

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES & INSTITUTE OF COMPUTATIONAL MATHEMATICS AND SCIENTIFIC/ENGINEERING COMPUTING, ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING, CHINA.
Email address: yeyuanda@lsec.ac.cc.cn